A Task-Centric Survey of Long-Document Visual Question Answering: Taxonomy, Evolution, and Future Directions

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Abstract

Visual document understanding is essential for acquiring knowledge from visually rich documents and is attracting growing research interest. Existing surveys emphasize architectures or technical approaches, with limited attention to how task bottlenecks shape technical choices and research priorities. We present a task-centric survey that decomposes long-document visual question answering (VQA) into four components: document parsing, evidence organization and retrieval, generator organization and learning, and evaluation. For each, we develop a fine-grained taxonomy and trace motivations, technical responses, and cross-component connections. We focus on explicit retrieval while covering direct reading. Our temporal analysis traces expanding priorities from multimodal content understanding to sufficient, high-fidelity evidence acquisition and sustained evidence management. We outline future directions in evidence sufficiency, verifiable and correctable understanding, and budget-aware coordination, with an outlook on continual document understanding.[1]

1 Introduction

Visually rich documents, such as research papers and financial reports, play a vital role in preserving and communicating knowledge through text, tables, charts, images, and their spatial and cross-modal relationships. Visual document understanding (VDU) interprets this content and structure, encompassing foundational tasks such as information extraction, as well as higher-level visual question answering (VQA) (Chen et al., 2024). While information extraction identifies target fields, entities, and relations, document VQA selects and integrates evidence to answer questions (Mathew et al., 2021).

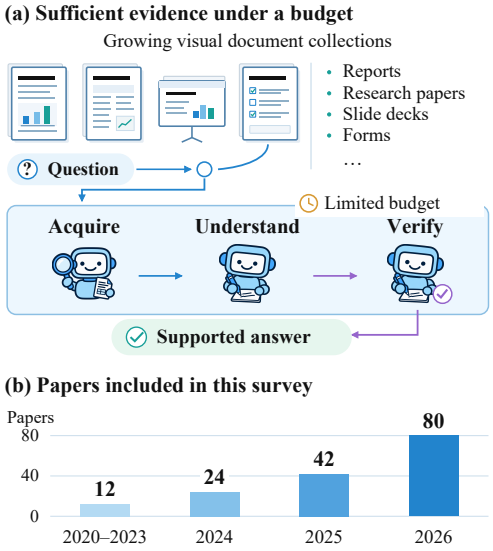

Figure 1: Core challenges of long-document VQA and publication trends of this field.

Document VQA has expanded from single-page understanding (Mathew et al., 2021) to evidence localization in MP-DocVQA (Tito et al., 2023), cross-page integration in SlideVQA (Tanaka et al., 2023), and long-context multimodal understanding in MMLongBench-Doc (Ma et al., 2024b). Longer documents and larger collections increase the computational costs and context demands of exhaustive reading, while textualization or visual compression may discard essential chart, layout, and fine-grained textual information (Faysse et al., 2025; Yu et al., 2025). Evidence dispersed across pages, sections, or documents requires recovering relationships and integrating complementary information (Tanaka et al., 2023; Cho et al., 2025). We focus on long-document VQA: how systems can acquire, understand, and verify sufficient evidence to answer questions within a limited budget (Fig. 1).

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[1]Project page: https://github.com/yufeiTongZJU/Long-Document-VQA-Survey.

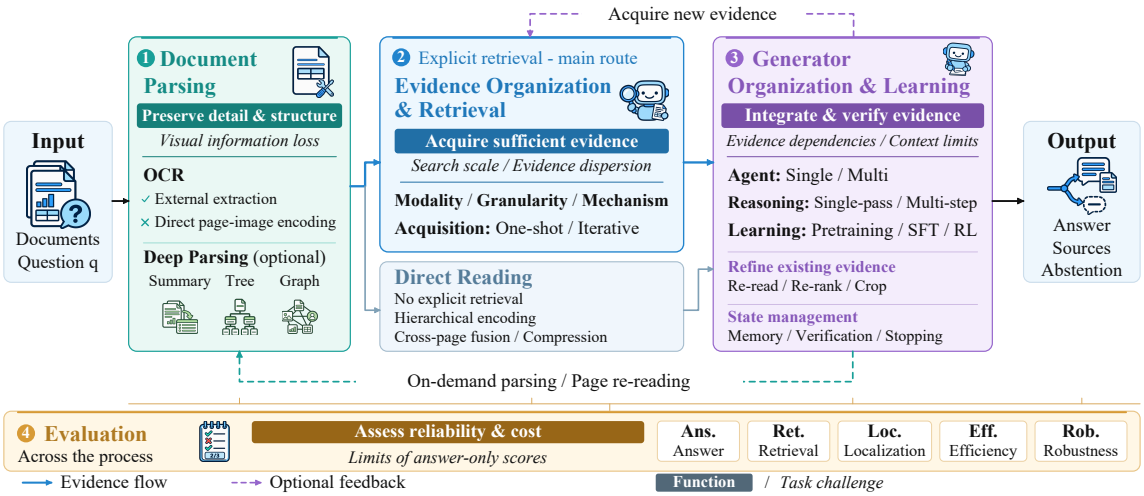


Figure 2: Task decomposition and two technical routes for long-document VQA. Colored bands indicate component functions, and italic labels identify the challenges they address. Both routes share parsing and generation, with evaluation throughout. Evidence sources and abstention are task-dependent outputs.

Advances in multimodal large language models (MLLMs) and demand for complex document understanding are accelerating document VQA research. Methods have expanded from multipage representations to visual retrieval-augmented generation (RAG) and feedback-driven evidence acquisition and reasoning (Tito et al., 2023; Yu et al., 2025; Wang et al., 2026d). Rapid literature growth since 2024 (Fig. 1) motivates a systematic review of methods, benchmarks, and technical evolution. Existing surveys cover document representations and multipage processing (Barboule et al., 2025), MLLM representations and training (Ding et al., 2026), multimodal retrieval (Gao et al., 2026), and visual retrievers and deployment (Yan et al., 2026c). These surveys primarily organize the literature by model architecture and technical approach. However, connections among long-document bottlenecks, fine-grained technical analysis, and evolving research motivations remain underexplored.

We analyze over 150 papers and organize this survey around task decomposition, fine-grained analysis, and temporal evolution. Our contributions are threefold: **(1) A unified, task-driven framework.** We map requirements for content fidelity, sufficient evidence acquisition, evidence use, and reliable evaluation onto four interconnected components: document parsing, evidence organization and retrieval, generator organization and learning, and evaluation. **(2) A fine-grained technical taxonomy by task function.** We categorize representation and structure, retrieval modalities and granularities, evidence acquisition strategies, reasoning organization and learning objectives, and evaluation capabilities. **(3) Technical evolution and future research directions.** We trace how task bottlenecks shape technical choices as research priorities expand from multimodal understanding to evidence acquisition and sustained management. We identify challenges and future directions in evidence sufficiency, verifiable and correctable understanding, and budget-aware coordination, with an outlook on continual document understanding.

2 Task Formulation and Decomposition

Task definition. Given a question q and a candidate set of visual documents $\mathcal{D}$, long-document VQA produces an answer under a resource budget B:

$$(\hat{a}, \hat{\mathcal{S}}) = F_\theta(q, \mathcal{D}; B), \tag{1}$$

where F_θ is the QA system, $\hat{a}$ its predicted answer, $\hat{\mathcal{S}}$ an optional set of evidence sources, and B the resource limits on context, computation, and tool calls. $\mathcal{D}$ contains one or more documents, each with ordered pages combining text, tables, charts, images, and layout information. In the *closed-domain* setting, the input document is specified for each question; in the *open-domain* setting, systems identify relevant sources from a shared document collection (Yu et al., 2025; Cho et al., 2025; Gao et al., 2026). Pages preserve context and provenance, elements provide finer-grained evidence, and cross-page or cross-document relations link evidence needed jointly.

Long documents pose three challenges. *(i) Input and search scale:* more candidate pages expand search spaces and increase context and computational demands for exhaustive reading. *(ii)*

Table 1: Representative explicit-retrieval methods. (YY.MM): first release. Within task groups, methods follow final conference submission deadlines, then first release within each conference; preprints/journal papers are inserted by first release. Venue omits tracks. Full details: Appendix C, Table 6.

Method		Parsing		Evidence Organization and Retrieval				Generation & Learning		
Name	**Venue**	**OCR**	**Deep**	**Modality**	**Gran.**	**Mechanism**	**Acquisition**	**Agent**	**Reasoning**	**Learning**
Retrieval-only methods										
DSE (2024a) *(24.06)*	EMNLP'24	✗	✗	Uni(V)	P	Visual^t	One-shot	–	–	–
ColPali (2025) *(24.06)*	ICLR'25	✗	✗	Uni(V)	P	Visual^t	One-shot	–	–	–
LMS-Retrieval (2026) *(26.08)*	ICDAR'26	✓	Tree	Uni(T)	E	Text^t	One-shot	–	–	–
CMDR-Embed (2026) *(26.07)*	ECCV'26	✗	✗	Uni(V)	P	Visual^t	One-shot	–	–	–
MMRetHeads (2026a) *(26.05)*	EMNLP'26	✗	✗	Uni(V)	P, E	Visual	One-shot	–	–	–
Document VQA methods										
MHST (2022) *(22.07)*	MM'22	✓	✗	Uni(T)	P, E	Text	One-shot	Single	Single-pass	SFT
CREAM (2024) *(24.10)*	MM'24	✓	✗	Uni(T)	P, E	Text + LLM	One-shot	Single	Single-pass	SFT
VisRAG (2025) *(24.10)*	ICLR'25	✗	✗	Uni(V)	P	Visual^t	One-shot	Single	Single-pass	✗
SV-RAG (2024) *(24.11)*	ICLR'25	✗	✗	Uni(V)	P	Visual^t	One-shot	Single	Single-pass	SFT
VisDoMRAG (2025) *(24.12)*	NAACL'25	✓	✗	Mixed(T/V)	P, E	Text + Visual	One-shot	Multi	Single-pass	✗
VDocRAG (2025) *(25.04)*	CVPR'25	✗	✗	Uni(V)	P	Visual^t	One-shot	Single	Single-pass	SFT
Doc-React (2025a) *(24.12)*	ACL'25	✗	✗	Uni(V)	P	Visual + VLM	Iterative	Single	Multi-step	✗
NeuSym-RAG (2025) *(25.05)*	ACL'25	✓	Graph, Summary	Mixed(T/V)	D, P, E	Text + MM + VLM	Iterative	Single	Multi-step	✗
MDocAgent (2025) *(25.03)*	arXiv'25	✓	✗	Mixed(T/V)	P	Text + Visual	One-shot	Multi	Single-pass	✗
VRAG-RL (2026d) *(25.05)*	NIPS'25	✗	✗	Uni(V)	P, E	Visual + VLM^t	Iterative	Single	Multi-step	SFT+RL
RA-Tuning (2025) *(24.11)*	EMNLP'25	✓	✗	Uni(V)	P, E	Visual	One-shot	Single	Single-pass	SFT
ViDoRAG (2025) *(25.02)*	EMNLP'25	✓	✗	Mixed(T/V)	P	Text + Visual + VLM	One-shot	Multi	Multi-step	✗
SimpleDoc (2025) *(25.06)*	EMNLP'25	✓	Summary	Mixed(T/V)	P	Visual + LLM	Iterative	Single	Multi-step	✗
MoLoRAG (2025b) *(25.09)*	EMNLP'25	✗	Graph	Uni(V)	P	Visual + VLM	One-shot	Single	Single-pass	✗
LILaC (2025) *(25.11)*	EMNLP'25	✓	Graph	Mixed(T/V)	E	MM	One-shot	Single	Single-pass	✗
DocAgent (2025a) *(25.11)*	EMNLP'25	✓	Tree	Uni(T)	P, E	Text + VLM	Iterative	Multi	Multi-step	✗
M3DocRAG (2025) *(24.11)*	ICCV'25	✗	✗	Uni(V)	P	Visual	One-shot	Single	Single-pass	✗
RegionRAG (2026e) *(25.10)*	AAAI'26	✗	✗	Uni(V)	E	Visual^t	One-shot	Single	Single-pass	✗
URaG (2026) *(25.11)*	AAAI'26	✗	✗	Uni(V)	P	Visual^t	One-shot	Single	Single-pass	SFT
EXTRACTOR (2026) *(25.09)*	ICLR'26	✓	Graph	Uni(T)	D, P, E	Text + LLM^t	Iterative	Single	Multi-step	SFT
SCoPE VLM (2026) *(25.10)*	EACL'26	✗	✗	Uni(V)	P	VLM^t	Iterative	Single	Multi-step	SFT+RL
Doc-Researcher (2026b) *(25.10)*	WWW'26	✓	Summary	Mixed(T/V)	D, P, E	MM + LLM	Iterative	Multi	Multi-step	✗
DMAP (2026) *(26.01)*	WWW'26	✓	Tree, Summary	Mixed(T/V)	P, E	Text + Visual + LLM	Iterative	Multi	Multi-step	✗
RAG-Anything (2025) *(25.10)*	arXiv'25	✓	Graph	Uni(T)	E	Text	One-shot	Single	Single-pass	✗
DocPrune (2026) *(26.04)*	CVPR'26	✗	✗	Uni(V)	P	Visual	One-shot	Single	Single-pass	✗
CogDoc (2025) *(25.12)*	arXiv'25	✗	✗	Uni(V)	P	VLM^t	One-shot	Single	Single-pass	RL
DocLens (2026a) *(25.11)*	ACL'26	✓	✗	Mixed(T/V)	P, E	VLM	One-shot	Multi	Single-pass	✗
Doc-V* (2026) *(26.04)*	ACL'26	✗	✗	Uni(V)	P	Visual + VLM^t	Iterative	Single	Multi-step	SFT+RL
MM-Doc-R1 (2026) *(26.04)*	ACL'26	✓	Tree	Uni(T)	P	Text	Iterative	Multi	Multi-step	RL
VimRAG (2026e) *(26.02)*	ICML'26	✗	✗	Uni(V)	P	MM + VLM^t	Iterative	Single	Multi-step	SFT+RL
MoDora (2026a) *(26.02)*	PACMMOD'26	✓	Tree, Summary	Mixed(T/V)	E	Text + LLM + VLM	One-shot	Single	Single-pass	✗
VisRAG 2.0 (2025b) *(25.10)*	MM'26	✗	✗	Uni(V)	P	Visual	One-shot	Single	Single-pass	SFT+RL
AgenticOCR (2026f) *(26.02)*	NIPS'26	✓	✗	Mixed(T/V)	P, E	VLM^t	One-shot	Single	Multi-step	SFT+RL
DocAtlas (2026b) *(26.07)*	NIPS'26	✓	Tree, Summary	Mixed(T/V)	P, E	LLM + VLM^t	Iterative	Single	Multi-step	RL
DocDancer (2026d) *(26.01)*	EMNLP'26	✓	Tree	Uni(T)	P, E	Text	Iterative	Single	Multi-step	SFT
AWM (2026a) *(26.08)*	EMNLP'26	✗	✗	Uni(V)	P	MM	Iterative	Single	Multi-step	SFT+RL
Doc-REFRAG (2026a) *(26.08)*	EMNLP'26	✗	✗	Uni(V)	P	MM	One-shot	Single	Single-pass	SFT+RL
EviProp (2026c) *(26.06)*	arXiv'26	✓	Graph	Mixed(T/V)	P	Visual + Text	One-shot	Single	Single-pass	✗
MAGE-RAG (2026) *(26.06)*	arXiv'26	✓	Graph, Summary	Mixed(T/V)	P, E	Visual + VLM	Iterative	Single	Multi-step	✗
HierDoc (2026) *(26.07)*	arXiv'26	✓	✗	Mixed(T/V)	P, E	VLM^t	One-shot	Single	Single-pass	✗
DocTrace (2026a) *(26.08)*	arXiv'26	✓	✗	Uni(V)	P, E	VLM^t	One-shot	Single	Single-pass	SFT+RL
DocMemo (2026b) *(26.08)*	arXiv'26	✓	Tree, Summary	Mixed(T/V)	P	Visual + LLM	Iterative	Multi	Multi-step	✗
D2-ScaleAgent (2026a) *(26.08)*	arXiv'26	✗	✗	Uni(V)	P, E	Visual + VLM	Iterative	Multi	Multi-step	✗
DocClaw (2026b) *(26.08)*	arXiv'26	✓	Tree	Mixed(T/V)	P, E	Text + Visual + VLM	Iterative	Single	Multi-step	✗
SCORE (2026c) *(26.09)*	arXiv'26	✗	✗	Uni(V)	P, E	Visual + VLM^t	Iterative	Single	Multi-step	SFT+RL

Notation: ✗ = absent; D/P/E = Document/Page/Element. Mechanism groups retriever families; + combines families; t marks added retrieval, scoring, or page-selection training for at least one component in the marked family. **Retriever families:** Visual = visual page/region retrieval; Text = text retrieval/reranking; MM = multimodal retrieval; LLM/VLM = language/vision-language models for selection, reranking, navigation, or region refinement.

Visual content complexity: information in small text, charts, and spatial layouts may be lost through resizing, conversion to text, or compression. *(iii) Evidence distribution and dependencies:* answers may require complementary evidence across pages or documents; local relevance does not ensure sufficiency (Ma et al., 2024b; Yu et al., 2025).

Four analytical components. Four interconnected components address these challenges. ***(1) Document parsing*** addresses visual information loss by preserving details and structure. Content extraction, visual preservation, and semantic structure construction provide readable representations and navigable structures, balancing fidelity against processing cost. ***(2) Evidence organization and retrieval*** addresses search scale and dispersion by acquiring sufficient evidence amid noise. It locates and selects evidence using document representations and structures, passing it and its sources to the generator. ***(3) Generator organization and learning*** addresses evidence dependencies and context limits through evidence integration and verification. It covers evidence presentation, organizational forms, reasoning processes, and learning objectives under computational constraints; intermediate judgments can guide acquisition, rereading, or verification. ***(4) Evaluation*** goes beyond answer scores to assess correctness, grounding, efficiency, and reliability throughout the process. Checks of evidence acquisition and use or controlled experiments diagnose parsing, retrieval, and reasoning errors and compare techniques' benefits and costs.

Table 2: Taxonomy of representative direct-reading methods. Full comparison: Appendix B, Table 5.

Method	Venue	OCR	Model	Learning
Single-page VQA				
Donut (2022) *(21.11)*	ECCV'22	✗	BART	PT+SFT
LayoutLMv3 (2022) *(22.04)*	MM'22	✓	RoBERTa	PT+SFT
Pix2Struct (2023) *(22.10)*	ICML'23	✗	Pix2Struct	PT+SFT
DocVLM* (2025) *(24.12)*	CVPR'25	✓	Qwen-VL	SFT
Multi-page VQA				
M3D (2023) *(23.01)*	AAAI'23	✓	T5	SFT
Hi-VT5 (2023) *(22.12)*	PR'23	✓	T5	PT+SFT
GRAM (2024) *(24.01)*	CVPR'24	✓	T5	SFT
mPLUG-DocOwl2 (2025) *(24.09)*	ACL'25	✗	Llama	PT+SFT
DocR1 (2026b) *(25.08)*	AAAI'26	✗	Qwen-VL	RL
DocSeeker (2026b) *(26.04)*	CVPR'26	✗	Qwen-VL	SFT+RL

All methods use a Single agent and no Deep Parsing. PT = Pretraining. Model lists backbone families only. DocVLM* primarily evaluated on single pages but also supports multiple pages.

Two technical routes. Fig. 2 connects these components through ***direct reading*** and ***explicit retrieval***. Direct reading answers from given document representations and may use hierarchical encoding, cross-page fusion, or compression to manage long-input costs (Tito et al., 2023). Explicit retrieval selects question-relevant documents, pages, or elements (e.g., text blocks, tables, and region crops) for the generator to read (Yu et al., 2025; Cho et al., 2025; Dong et al., 2026b). Reasoning feedback can guide further evidence acquisition (Wang et al., 2026d), local rereading, or on-demand parsing of existing evidence. These components describe task functions; their implementation and execution order may vary. Explicit retrieval accounts for approximately 80% of surveyed methods, motivating our focus on this route while also covering direct reading. Sec. 3 develops a fine-grained taxonomy by task function and traces each component's research motivations and technical evolution.

3 Taxonomy and Technical Evolution

For each component, we develop a fine-grained taxonomy, compare technical benefits and costs, and trace evolving research motivations and technical responses. Stages reflect expanding research priorities: earlier approaches coexist, and methods may adopt techniques from different stages across components. Tables 1 and 2 compare representative explicit-retrieval and direct-reading methods; full comparisons appear in Appendices B and C.

3.1 Document Parsing

Technical taxonomy. Document parsing converts raw pages into accessible content and structure. Tables 1 and 2 distinguish two independent dimensions: *OCR* and *Deep Parsing*. OCR (✓) denotes dependence on external extraction or textualization, including OCR, PDF text extraction, and VLM transcription or description of charts and tables. Extracted text supports indexing and language processing, but linearization may lose layout, text–image correspondences, and table structure, while recognition errors can propagate downstream (Ma et al., 2024a; Faysse et al., 2025; Yu et al., 2025; Chen et al., 2024). OCR (✗) methods primarily encode pages visually to reduce textualization loss but remain limited by resolution and visual reading ability (Yu et al., 2025; Chen et al., 2024). Methods combining extracted or transcribed text with page images are marked OCR (✓) (Suri et al., 2025; Han et al., 2025). Deep Parsing (✓) builds semantic overviews or explicit relations: *Summary* captures page or document semantics, *Tree* represents section hierarchies and containment, and *Graph* organizes text–image, entity, or cross-page relations (Jain et al., 2025; Wu et al., 2025b; Yun et al., 2025; Sun et al., 2025a); these structures can be combined. Element transcription or description alone does not count as Deep Parsing. Parsing may cover full documents in advance or proceed on demand, guided by the question or ongoing process.

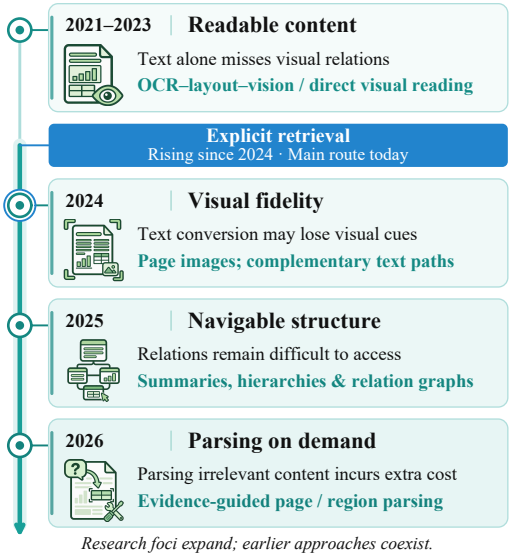


Figure 3: Four stages of document parsing: bottlenecks in black, technical responses in teal.

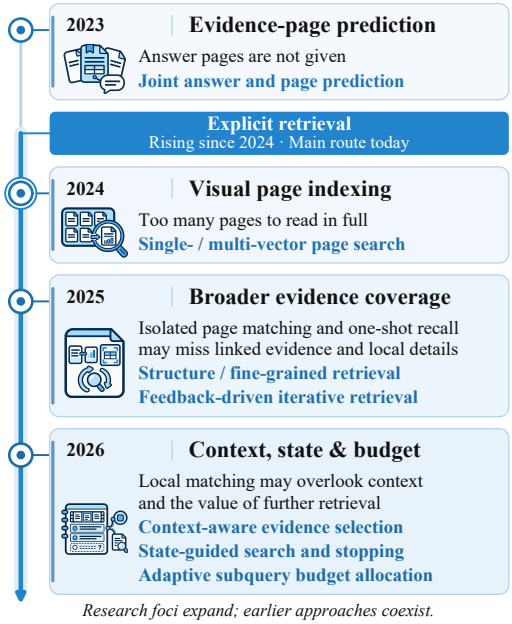
Evolution of research focus (Fig. 3). Early work prioritized *readable content* while preserving layout and visual relations. LayoutLMv3 and UDOP jointly model OCR text, layout, and images (Huang et al., 2022; Tang et al., 2023), while Donut, Pix2Struct, and UReader learn directly from pages or screenshots (Kim et al., 2022; Lee et al., 2023; Ye et al., 2023). From 2024, explicit retrieval emphasized *original-page visual fidelity*: DSE, ColPali, VisRAG, and SV-RAG use page images di-

rectly (Ma et al., 2024a; Faysse et al., 2025; Yu et al., 2025; Chen et al., 2024), while VisDoMRAG and MDocAgent combine visual and textual paths to mitigate omissions and reading biases (Suri et al., 2025; Han et al., 2025). In 2025, *navigable structure* made content relations accessible: SimpleDoc and Doc-Researcher summarize pages and documents, respectively (Jain et al., 2025; Dong et al., 2026b); DocAgent builds section hierarchies (Sun et al., 2025a); and MoLoRAG and LILaC construct page-similarity and layered component graphs, respectively (Wu et al., 2025b; Yun et al., 2025). To avoid parsing irrelevant content at scale, work in 2026 explores *parsing on demand*: AgenticOCR performs question-guided regional OCR (Wang et al., 2026f); DocLens, HierDoc, and DocTrace extract elements or parse selected pages (Zhu et al., 2026a; Gu et al., 2026; Xiang et al., 2026a); and DocClaw adapts parsing tool use to tasks and intermediate results (Xiang et al., 2026b).

Parsing supports evidence acquisition (Sec. 3.2); its value should be assessed by evidence completeness, answer quality, and cost, rather than transcription accuracy or structural richness alone.

3.2 Evidence Organization and Retrieval

Technical taxonomy. Evidence organization and retrieval select and assemble evidence from parsed content and structures, but local relevance does not ensure sufficient support for cross-page reasoning (Cho et al., 2025). Table 1 compares four dimensions. *Modality* specifies evidence-side matching representations: Uni(T) for text, Uni(V) for vision, and Mixed(T/V) for both. *Granularity* denotes evidence units supplied to generators or returned by retrieval-only methods: Document covers document-level evidence; Page covers full-page images or complete parsed page content; Element covers independently accessible text blocks, sections, charts, tables, or region crops. Pages preserve context but may add noise, whereas elements emphasize details but risk losing context. Modality or granularity may change before generation, as when ViDoRAG maps text chunks to source page images (Wang et al., 2025); Table 6 details these mappings. *Mechanism* includes lexical or unimodal/multimodal vector matching and LLM/VLM-based page selection, reranking, or region selection, optionally combined with tree navigation or graph expansion. The table lists retriever families. *Acquisition* distinguishes feedback-driven expansion: Iterative uses generation or control feedback to acquire new evidence beyond the current set; One-shot permits fixed multistage retrieval, reranking, or refinement without such expansion.



Research foci expand; earlier approaches coexist.

Figure 4: Four stages of evidence organization and retrieval: bottlenecks in black, technical responses in blue.

Evolution of research focus (Fig. 4). Hi-VT5 and M3D jointly learn QA and evidence-page prediction over given multipage inputs (Tanaka et al., 2023; Tito et al., 2023), providing a precursor to explicit retrieval. In 2024, input capacity limits motivated *visual page indexing* and the rise of explicit retrieval. DSE uses single-vector screenshot embeddings (Ma et al., 2024a); ColPali uses multivector late interaction (Faysse et al., 2025); and VisRAG, SV-RAG, and M3DocRAG connect page retrieval to visual QA (Yu et al., 2025; Chen et al., 2024; Cho et al., 2025). In 2025, incomplete coverage of linked evidence and local details motivated *structural relations, fine-grained acquisition, and feedback-driven iterative retrieval*. MoLoRAG and LILaC link complementary evidence through page or component relations (Wu et al., 2025b; Yun et al., 2025); RegionRAG and Doc-Researcher support regional and multigranular acquisition, respectively (Li et al., 2026e; Dong et al., 2026b); and VRAG-RL updates queries, retrieves additional pages, and refines acquired image regions during reasoning (Wang et al., 2026d). In 2026, attention expands to *context, explicit evidence states, and budget constraints*. Doc-V* uses document-wide thumbnails for page navigation (Zheng et al., 2026), while CMDR-Embed incorporates multipage context into page representations (Tanaka et al., 2026). MAGE-RAG tracks page and element states (Zuo et al.,

2026); DocAtlas updates retrieval structures with reading-derived evidence (Wei et al., 2026b); and DocMemo updates page beliefs and question memory to guide search, reading, and stopping (Yao et al., 2026b). MAB-DQA allocates retrieval budgets by estimating each subquestion's acquisition value from VLM feedback (Xiang et al., 2026c).

More evidence may add noise and redundancy without ensuring sufficient support. Sec. 3.3 examines how generators integrate and verify evidence and guide further acquisition through feedback.

3.3 Generation and Learning

Technical taxonomy. Generation and learning produce grounded answers under context and computational budgets. Reading all pages preserves context at increasing input cost, whereas reading retrieved evidence narrows inputs but must address evidence gaps and noise. Tables 1 and 2 compare explicit-retrieval and direct-reading methods. *Agent* distinguishes Single and Multi by runtime reasoning roles and collaboration. *Reasoning* distinguishes fixed workflows from feedback loops: Single-pass covers a single chain of thought, structured decoding, or a fixed multi-agent pipeline; Multi-step revises judgments and actions through repeated feedback from intermediate results, tool observations, or verification. Multi-step may acquire new evidence through feedback (Iterative acquisition) or only reread and verify existing evidence, as in ViDoRAG's One-shot acquisition (Wang et al., 2025). *Learning* records added generator-side training: Pretraining develops document representations and cross-modal alignment; SFT uses answer, evidence, or action demonstrations; RL optimizes policies through answer, evidence, or process rewards.

Evolution of research focus (Fig. 5). In 2021–2023, pretraining and task fine-tuning established document understanding and grounded answering (Tanaka et al., 2023; Tito et al., 2023; Kim et al., 2022; Lee et al., 2023). In 2024, growing page and token counts motivated *long-input handling*: GRAM uses document tokens for cross-page communication (Blau et al., 2024), while mPLUG-DocOwl2 compresses high-resolution visual features (Hu et al., 2025). VisRAG and SV-RAG read retrieved pages (Yu et al., 2025; Chen et al., 2024), while Retrieval-aware Tuning fine-tunes generators with evidence and distractor pages (Chia et al., 2025). In 2025, single-pass omissions and misreadings motivated *evidence-guided learning, feedback reasoning, and tool actions*. ViDoRAG uses

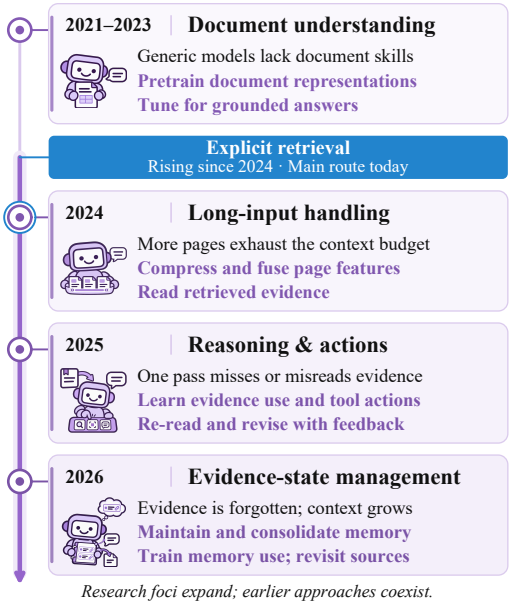


Figure 5: Four stages of generation and learning: bottlenecks in black, technical responses in purple.

cross-role feedback to reread existing candidates (Wang et al., 2025); DocR1 improves localization and answering with evidence-page rewards (Xiong et al., 2026b); and VRAG-RL uses SFT and RL to train VLMs to alternate among search, regional inspection, and answering (Wang et al., 2026d). In 2026, feedback reasoning incorporates *evidence-state management* to reduce forgetting and context accumulation. DocMemo provides evidence and question memory (Yao et al., 2026b); AWM rewards memory maintenance for answering (Zhou et al., 2026a); and SCORE retains provenance, reloads source pages, and reorganizes context before answering (Shen et al., 2026c).

Within feedback loops, Sec. 3.2 examines how evidence states guide acquisition; this section covers evidence integration, answer verification, and feedback on remaining gaps. Organization and learning should be evaluated jointly by answer quality, evidence support, and end-to-end cost.

3.4 Benchmarks and Evaluation

Evaluation dimensions. Table 3 compares benchmarks by scale, evidence requirements, and evaluation capabilities. *Granularity* denotes the verification level supported by public evidence annotations; *Evaluation* identifies assessable capabilities. Ans. measures answer correctness; Ret. evaluates evidence recall and ranking; Loc. measures spatial alignment between predicted and annotated regions. Eff. covers parsing, indexing, and online inference costs; Rob. assesses performance under

Table 3: Representative benchmarks grouped by evaluation goal and input scale. Full details: Appendix A, Table 4.

Benchmark			Data Size		Question Type				Evidence and Evaluation		
Name	**Venue**	**Domain**	**#Pages**	**#QA**	**MD**	**MP**	**SP**	**Unans.**	**Source**	**Gran.**	**Evaluation**
Retrieval-focused benchmarks											
ViDoRe (2025) *(24.06)*	ICLR'25	Open	1.0k	3.81k	✗	✗	✓	✗	TXT/L/C/TAB/I	P	Ans. Ret.
REAL-MM-RAG (2025) *(25.02)*	ACL'25	Open	2.1k	4.6k	✗	✗	✓	✗	TXT/C/TAB/I	P	Ans. Ret.
ViDoRe V2 (2025) *(25.03)*	arXiv'25	Open	1.1k	327	✓	✓	✓	✗	TXT/L/C/TAB/I	P	Ret.
MMDocIR (2025) *(25.01)*	EMNLP'25	Closed	65.1	1.66k	✗	✓(18.9%)	✓(81.1%)	✗	TXT/L/C/TAB/I	P, E	Ret. Loc.
CMDR-Bench (2026) *(26.07)*	ECCV'26	Closed	183.5	800	✗	✓	–	✗	TXT/L/C/TAB/I	P	Ret.
Answer-quality benchmarks: single-page inputs											
DocVQA (2021) *(20.07)*	WACV'21	Closed	1.0	50k	✗	✗	✓	✗	TXT/L/C/TAB/I	–	Ans.
InfographicVQA (2022) *(21.04)*	WACV'22	Closed	1.0	30k	✗	✗	✓	✗	TXT/L/C/TAB/I	–	Ans.
ChartQA (2022) *(22.03)*	ACL'22	Closed	1.0	32.7k	✗	✗	✓	✗	C	–	Ans.
DOGR-Bench (2025) *(24.11)*	ICCV'25	Closed	1.0	3.6k	✗	✗	✓	✗	TXT/L/C/I	E	Ans. Loc.
Answer-quality benchmarks: multipage inputs or input size unreported											
TAT-DQA (2022) *(22.07)*	MM'22	Closed	1.1	16.6k	✗	✓	✓	✗	TXT/L/TAB	–	Ans.
SlideVQA (2023) *(23.01)*	AAAI'23	Closed	20.0	14.5k	✗	✓(13.9%)	✓(86.1%)	✗	TXT/L/C/TAB/I	P	Ans. Ret.
MP-DocVQA (2023) *(22.12)*	PR'23	Closed	8.3	46.2k	✗	✗	✓	✗	TXT/L/C/TAB/I	P	Ans. Ret.
DUDE (2023) *(23.05)*	ICCV'23	Closed	5.72	41.5k	✗	✓(2.1%)	✓(85.2%)	✓(12.7%)	TXT/L/C/TAB/I	–	Ans.
MMLongBench-Doc (2024b) *(24.07)*	NIPS'24	Closed	47.5	1.1k	✗	✓(33.7%)	✓(45.7%)	✓(20.6%)	TXT/L/C/TAB/I	P	Ans. Ret.
Wiki-/Paper-VISA (2025a) *(24.12)*	ACL'25	Open	–	90k; 102.2k	✗	✗	✓	✓	TXT/TAB/I	P, E	Ans. Ret. Loc.
M-LongDoc (2025) *(24.11)*	EMNLP'25	Closed	210.8	0.9k	✗	✗	✓	✗	TXT/C/TAB/I	P	Ans. Ret.
DVisRAG (2026a) *(26.02)*	CVPR'26	Open	–	367.6k	✗	✓	✓	✗	TXT/L/C/TAB/I	P	Ans. Ret. Rob.
ViDoRe V3 (2026) *(26.01)*	ACL'26	Open	–	3.1k	✓	✓	✓	–	TXT/L/C/TAB/I	P, E	Ans. Ret. Loc.
SynthDocBench (2026) *(26.07)*	COLM'26	Closed	51.1	1.8k	✗	✓	✓	✗	TXT/C/TAB	E	Ans. Ret.
DocScope (2026) *(26.05)*	NIPS'26	Closed	51.3	1.12k	✗	✓(57.7%)	✓(35.3%)	✓(6.9%)	TXT/C/TAB/I/L	P, E	Ans. Ret. Loc.

Open/Closed: cross-document corpus/given document. #Pages: mean input pages (document mean if per-question statistics are unavailable); Open uses the full corpus. #QA: questions/queries; –: unreported/inapplicable. MD/MP/SP: multi-document/multi-page/single-page evidence; Unans.: unanswerable. TXT/L/C/TAB/I: text/layout/charts/tables/images; P/E: Page/Element.

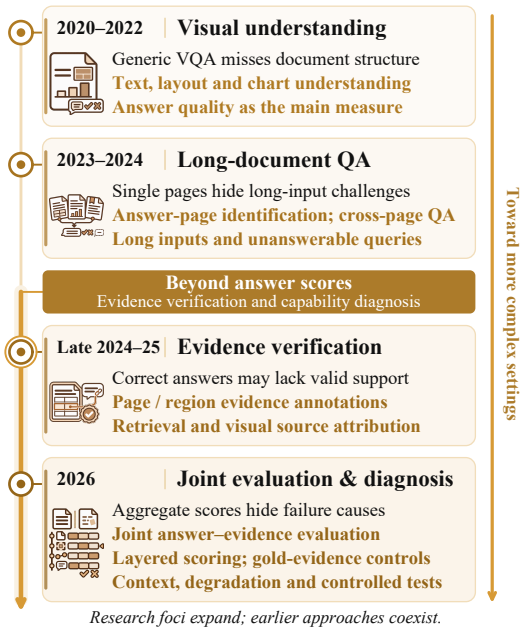


Figure 6: Four stages of benchmarks and evaluation: bottlenecks in black, technical responses in amber.

visual degradation or interference. Joint reporting under comparable inputs and budgets helps relate capability gains to end-to-end costs.

Evolution of research focus (Fig. 6). In 2020–2022, DocVQA, InfographicVQA, and ChartQA benchmarked text, layout, and chart understanding (Mathew et al., 2021, 2022; Masry et al., 2022). In 2023–2024, longer inputs shifted evaluation toward page identification and cross-page integration (Tanaka et al., 2023; Tito et al., 2023). DUDE adds complex and unanswerable questions (Van Landeghem et al., 2023), while MMLongBench-Doc and M-LongDoc assess long-document multimodal understanding (Ma et al., 2024b; Chia et al., 2025). From late 2024 through 2025, RAG prompted evidence and provenance checks beyond answer scores: VISA grounds answers in source-page regions (Ma et al., 2025a), while MMDocIR evaluates page- and element-level retrieval (Dong et al., 2025). Work in 2026 targets *joint evaluation and capability diagnosis in complex settings*. ViDoRe V3 links retrieval and QA on a shared corpus, with annotated relevant pages as a control (Loison et al., 2026); DocScope diagnoses bottlenecks through page, region, fact, and answer scores under varying evidence conditions (Feng et al., 2026). CMDR-Bench requires cross-page context for retrieval (Tanaka et al., 2026); DVisRAG evaluates retrieval, generation, and end-to-end robustness under visual degradation (Chen et al., 2026a); and SynthDocBench uses controlled synthesis and structured evidence metadata to diagnose capabilities across document conditions and question types (Verma et al., 2026). Across stages, tasks expand to longer documents, cross-page and multi-element evidence, multihop reasoning, and diverse applications.

4 Research Trends and Future Directions

Building on component analyses, we examine how their interactions shape priorities and outline challenges and future directions at two levels: improving individual-question answering and extending the task toward continual document understanding.

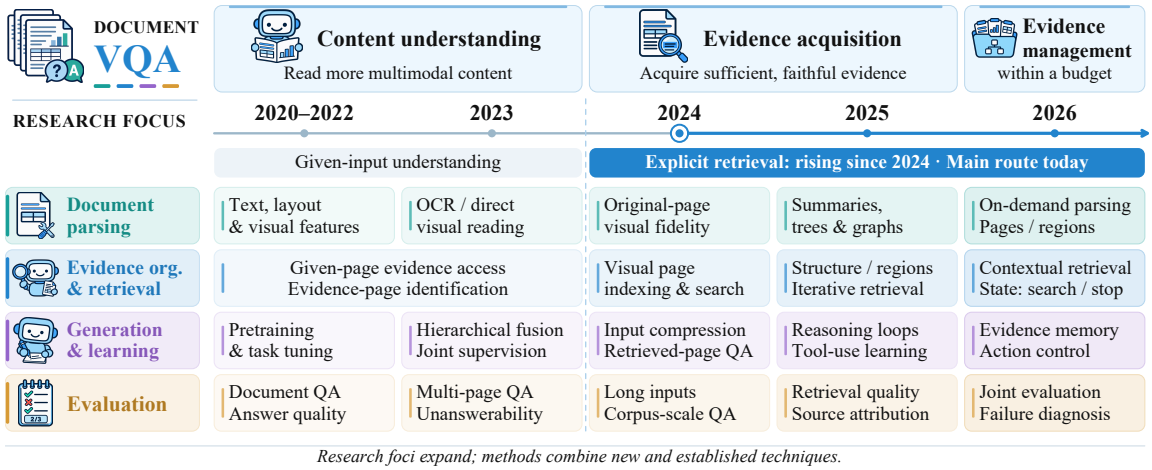


Figure 7: Research foci and technical evolution across four components of document VQA. Five periods span three overarching foci: content understanding, evidence acquisition, and sustained evidence management.

4.1 Cross Component Evolution

Document VQA research has broadened from content understanding to evidence acquisition and sustained evidence management (Fig. 7). In 2020–2023, multipage representations and cross-page fusion extended single-page understanding (Tanaka et al., 2023; Tito et al., 2023). In 2024–2025, costly exhaustive reading motivated sufficient, high-fidelity evidence acquisition. Visual page indexing and reading after retrieval in 2024 helped establish explicit retrieval as the main route (Faysse et al., 2025; Yu et al., 2025). Structural relations, fine-grained selection, and feedback-driven acquisition improved coverage and use of local details in 2025 (Wang et al., 2026d; Yun et al., 2025). In 2026, attention extends to maintaining evidence states, identifying gaps, and coordinating evidence use and verification across rounds under budget constraints (Zhou et al., 2026a; Yao et al., 2026b; Shen et al., 2026c).

Bottlenecks also propagate across components. Retrieval narrows inputs but risks omissions, motivating structure-aware acquisition and on-demand parsing to preserve evidence relations and details (Yun et al., 2025; Wang et al., 2026f). Feedback makes parsing and retrieval schedulable actions, with generation maintaining evidence states and guiding rereading, acquisition, and stopping (Wang et al., 2026d; Yao et al., 2026b). Evaluation must diagnose failures through answers and evidence (Feng et al., 2026) and assess gains against end-to-end cost. Methods may update selected components while retaining established techniques elsewhere; evolution thus reflects broadening priorities rather than wholesale replacement.

4.2 Challenges and Future Directions

For individual questions, we identify three research directions. **Evidence sufficiency.** Cross-page context and region extraction improve acquisition (Tanaka et al., 2026; Wang et al., 2026f), but local relevance and precise localization do not ensure sufficient support. Future work should construct question-driven evidence sets and assess their sufficiency. **Verifiable and correctable document understanding.** Cross-round memory and multi-level evaluation support process verification (Feng et al., 2026; Yao et al., 2026b), but correct answers may mask unreliable intermediate judgments, and early errors can propagate. Future work should link source tracing, process verification, and error correction. **Budget-aware coordination.** Action learning and dynamic retrieval improve adaptive acquisition (Wang et al., 2026d), but component gains may not improve the full pipeline. Future work should coordinate decisions across components using evidence states and budget constraints to balance answer quality and end-to-end cost.

Document use often involves follow-up questions and reading across tasks, motivating **memory reuse across questions** and **continual, experience-driven strategy improvement**. The former reuses evidence and interaction states for subsequent questions; the latter turns cross-task experience into transferable retrieval, reading, and verification strategies. Appendix D details challenges, research opportunities, and task extensions.

5 Conclusion

This survey decomposes long-document VQA into four components and links a fine-grained taxon-

omy with task bottlenecks, research motivations, and technical evolution. Emphasizing explicit retrieval while covering direct reading, we trace expanding priorities from multimodal understanding to sufficient, high-fidelity evidence acquisition and sustained management under limited budgets. Future directions include evidence sufficiency assessment, verifiable and correctable understanding, and budget-aware coordination for individual questions, alongside memory reuse across questions and strategy improvement across tasks toward continual document understanding.

Limitations

This survey focuses on document VQA as a higher-level understanding task, particularly evidence acquisition and use in long documents. Foundational tasks such as text and layout recognition and field and entity extraction are covered only where they affect question answering, without being reviewed in detail.

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A Benchmark Catalog

Beyond the representative comparison in Table 3, Table 4 catalogs all 55 benchmark records covered in this survey, retaining their scale, question types, and evaluation capabilities. Benchmarks are grouped by their primary evaluation goal into retrieval-oriented and answer-quality-oriented benchmarks, with the latter further grouped by input page count. Benchmark selection should distinguish input scale, evidence span, and the stage at which processing begins. Longer inputs do not necessarily require cross-page evidence combination: MP-DocVQA identifies single-page evidence among multiple candidate pages (Tito et al., 2023), whereas SlideVQA includes questions requiring evidence across pages (Tanaka et al., 2023). An open candidate corpus and a given document reflect different task settings, so page counts alone do not establish relative difficulty. Benchmarks such as SPIQA provide pre-extracted elements (Pramanick et al., 2024); their results primarily measure understanding and evidence selection over the supplied material, rather than full-page parsing. Evidence annotation granularity further determines what can be verified: document or page retrieval, region localization, and answer quality should be interpreted separately.

MMLongBench-Doc and M-LongDoc assess long-input processing (Ma et al., 2024b; Chia et al., 2025); SlideVQA and CMDR-Bench probe cross-page evidence requirements (Tanaka et al., 2023, 2026); and MMDocIR and VISA support fine-grained acquisition and source verification (Dong et al., 2025; Ma et al., 2025a). DVisRAG evaluates performance under visual degradation (Chen et al., 2026a), while DocScope analyzes answering bottlenecks under different evidence conditions (Feng et al., 2026). Cost comparisons should specify input conditions and budgets and account for parsing, indexing, and online inference, rather than assessing system capabilities through answer scores alone.

B Methods without Explicit Retrieval

Table 5 extends the direct-reading comparison in Table 2. These methods organize reading within the supplied content without explicit evidence retrieval in the answering pipeline, and are grouped into single-page and multipage methods according to their primary design and evaluation settings. Multipage methods make different trade-offs among information organization, input scale, and evidence use. Hi-VT5 organizes multipage inputs through hierarchical representations (Tito et al., 2023), GRAM uses document tokens for cross-page communication (Blau et al., 2024), and mPLUG-DocOwl2 compresses high-resolution visual features to control input scale (Hu et al., 2025). M3D and DocR1 strengthen the link between evidence identification and answering through evidence-and-answer supervision and evidence-page rewards, respectively (Tanaka et al., 2023; Xiong et al., 2026b). These designs act on representations, fusion, or learning objectives; Table 5 compares external extraction dependence, generation organization, backbone models, added training, and the dimensions actually evaluated.

Direct reading allocates resources among input capacity, detail fidelity, and cross-page relations. Compression reduces representation size but may lose critical text or visual details. More input pages broaden coverage and higher resolution improves detail visibility, but both increase reading cost. Comparisons should therefore consider answer quality, evidence selection, and cost together, distinguishing model capabilities from input configurations. Direct reading and explicit retrieval are complementary: hierarchical fusion and compression support multipage reading after retrieval, while retrieval further controls the evidence passed to generation.

C Methods with Explicit Retrieval

Table 6 catalogs 110 explicit-retrieval methods, extending the representative comparison in Table 1. For retrieval-only methods, the comparison covers evidence representations, matching, and returned results; generation attributes do not apply. For QA methods, it additionally records how evidence reaches the generator, agent organization, reasoning feedback, and added training. Parsing, retrieval, generation, and evaluation attributes should be read together: the same visual retriever can support different generation pipelines, and neither structural parsing nor multi-agent organization alone determines whether acquisition is iterative.

Retriever matching representations should be distinguished from generator inputs. *Modality* denotes the evidence-side modalities used to assess relevance, whereas *Granularity* denotes the evidence units actually received by the generator; retrieval-only methods are labeled by their returned units.

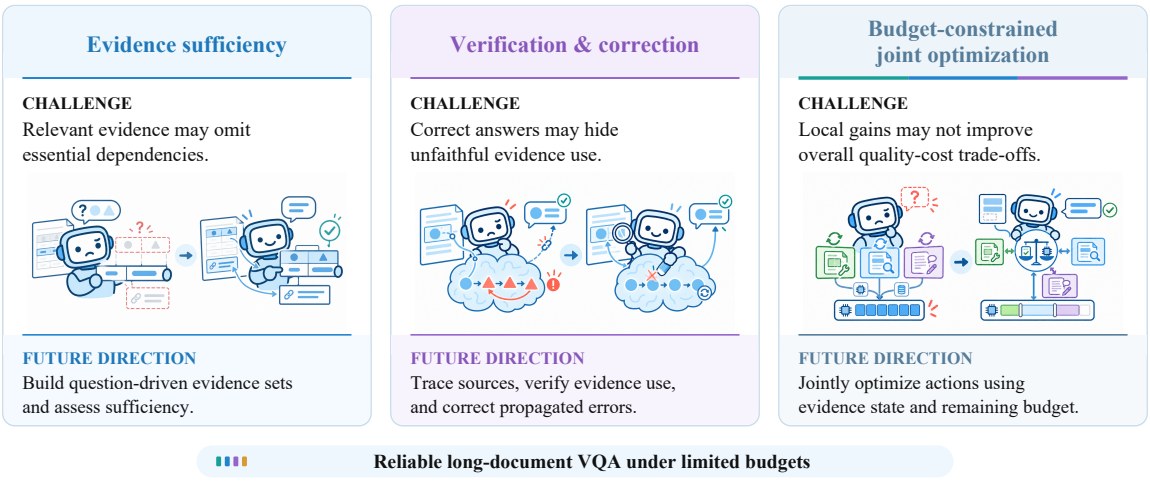

Figure 8: Challenges and future directions for answering individual questions in long-document VQA. The three panels correspond to Secs. D.1–D.3, covering evidence sufficiency, verifiable and correctable understanding, and budget-aware coordination toward reliable answers under limited budgets.

When retrieval hits and generation inputs differ in modality or granularity, *Mechanism* records the mapping with an arrow. For example, ViDoRAG maps retrieved text chunks to their source page images (Wang et al., 2025), allowing text matching to support visual page reading. *Iterative* acquisition introduces evidence beyond the current set in response to generation or control feedback; graph traversal without such feedback, fixed multistage retrieval, and reranking or refinement of existing evidence do not qualify. *Reasoning* instead describes whether generation involves a feedback loop. Thus, One-shot acquisition can coexist with Multi-step reasoning, whereas a single CoT or a fixed multi-agent pipeline is classified as Single-pass.

Superscript t in *Mechanism* marks added retrieval, scoring, or page-selection training. Generation-side *Learning* records added training of answering, control, or tool components; it does not imply that all modules are updated together. For example, MM-Doc-R1 and DocDancer train only the language agent while freezing the visual Read tool (Lin et al., 2026; Zhang et al., 2026d); AgenticOCR trains only the region tool agent while freezing the final answerer (Wang et al., 2026f); Doc-REFRAG applies RL only to its token-recovery module (Hu et al., 2026a); and RobustVisRAG performs feature-alignment training only on the generation-side visual encoder while freezing the language backbone (Chen et al., 2026a).

D Challenges and Research Opportunities

Building on Sec. 4.2, this appendix discusses challenges and research opportunities at two levels: improving capabilities within the current task and extending its boundaries. Secs. D.1–D.3 examine evidence sufficiency, verifiable and correctable understanding, and budget-aware coordination for individual questions, drawing on existing advances and limitations to identify future directions and concrete research opportunities (Fig. 8). Sec. D.4 extends the independent-question setting through memory reuse across questions and strategy improvement across tasks, broadening the perspective to sustained interaction and accumulated experience.

D.1 Evidence Sufficiency

Task-oriented evidence sufficiency moves beyond local relevance to the support required for an answer, asking: *Is the current evidence sufficient, and what is missing?* CMDR-Embed incorporates cross-page context into retrieval representations (Tanaka et al., 2026), while AgenticOCR narrows page reading to question-relevant regions (Wang et al., 2026f). Yet contextual association and fine-grained extraction can still omit essential dependencies. AgenticOCR explicitly notes that extracting only a table row or cell without its header and surrounding context can leave evidence incomplete. Finding relevant content therefore does not ensure sufficient support for answering.

Future direction. Move from matching individual evidence units by relevance to question-driven evidence set construction and sufficiency assessment. This requires identifying the facts needed for an answer, their dependencies, and the remaining gaps, providing testable grounds for adaptive acquisition and stopping.

Research opportunities. *(a) Model evidence requirements and dependencies.* Organize evidence around the facts, comparisons, and calculation conditions required by the question, distinguishing context used for localization from evidence that directly supports the answer. *(b) Select complementary evidence jointly.* Treat tables continued across pages, headers and values, or figures and their annotations as interdependent evidence combinations; balance coverage against redundancy and avoid fragmentation that breaks essential semantic dependencies. *(c) Learn testable sufficiency judgments.* Construct examples with complete evidence, partially missing evidence, and highly relevant but insufficient evidence to test whether models can identify gaps and translate them into explicit acquisition needs.

D.2 Verifiable and Correctable Understanding

Verifiable and correctable document understanding moves beyond correct answers to trustworthy evidence chains, asking: *Is acquired information used faithfully, and can errors be corrected?* Methods such as DocMemo explicitly maintain evidence states across rounds (Yao et al., 2026b), while DocScope verifies pages, regions, facts, and answers (Feng et al., 2026). DocScope's experiments show that correct answers can coexist with incomplete evidence chains. This raises a further question: can parsing or summarization errors enter memory, affect subsequent acquisition, and become repeatedly reinforced?

Future direction. Connect source tracing, process verification, and error correction to maintain traceable, correctable evidence states across rounds, and evaluate whether systems can detect, locate, and correct errors.

Research opportunities. *(a) Build revisable evidence memory.* Retain source pages, regions, and dependencies for intermediate facts and summaries; when misreadings or conflicts emerge, reread the sources, retract erroneous records, and update dependent judgments. *(b) Test whether evidence supports the answer.* Remove or replace key evidence under controlled conditions to test whether models detect missing support or changed facts, while checking stability under changes to irrelevant or redundant evidence. Distinguish citation accuracy, accurate fact reading, and evidential support for the answer. *(c) Connect diagnosis to correction.* Under cross-page dependencies, distractors, visual degradation, and unanswerable conditions, evaluate whether systems choose to reparse, acquire more evidence, revise the answer, or abstain, and measure their ability to recover from errors.

D.3 Decision Making under Budget Constraints

Budget-aware coordination moves from local optimization to end-to-end quality–cost trade-offs, asking: *Where should limited computation be spent?* Visual compression, tool-action learning in VRAG-RL (Wang et al., 2026d), on-demand parsing in AgenticOCR (Wang et al., 2026f), and memory-guided retrieval in DocMemo (Yao et al., 2026b) provide several forms of dynamic processing. However, savings in individual operations may be partly offset by auxiliary model calls, multiple interaction rounds, and state maintenance. Their quality gains and computational costs must therefore be assessed across the full pipeline.

Future direction. Building on dynamic processing and cross-component interactions, use evidence states and remaining budgets to jointly decide whether to search further, inspect local details, reparse, verify a judgment, or finish answering. Learn evidence acquisition and use jointly under a quality–cost objective.

Research opportunities. *(a) Allocate budgets dynamically across components.* Adapt parsing scope, visual resolution, retrieval depth, and reasoning rounds to question difficulty and current evidence gaps; determine when direct reading is preferable and when retrieval or tools are worthwhile. *(b) Learn the marginal value of actions.* Extend action learning by comparing how much an additional retrieval, local rereading, or verification step improves evidence coverage and answer quality, and incorporate cost into action selection and stopping policies. *(c) Test generalization across budgets and settings.* Compare quality–cost curves across document lengths, degrees of evidence dispersion, and computational budgets. Include preprocessing, indexing, auxiliary models, and online inference to assess whether policies adapt to practical resource constraints.

D.4 Towards Continual Document Understanding

The preceding three directions focus on answering individual questions. Building on them, this survey considers how to extend document VQA beyond independent questions. Practical document

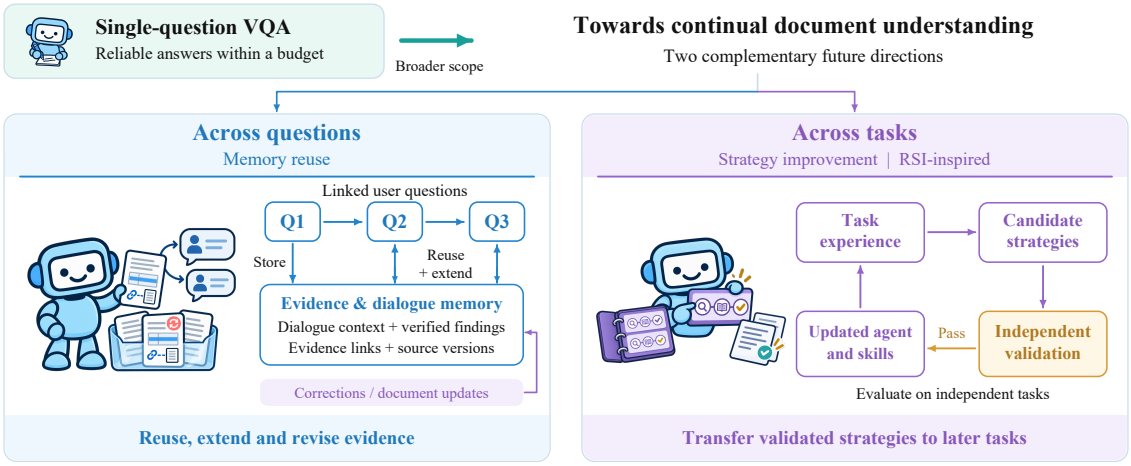

Figure 9: Two complementary directions from individual question answering toward continual document understanding. Across questions, systems reuse, supplement, and revise evidence and interaction memory. Across tasks, they propose strategies from experience, validate them on independent tasks before updating agents and skills, and continue accumulating experience.

use requires systems to solve the current question, exploit connections across successive interactions, and improve subsequent processing strategies through experience across tasks. We therefore identify two related extensions—memory reuse across questions and continual, experience-driven strategy improvement—that shift the perspective from individual answers to sustained interaction and accumulated capabilities (Fig. 9).

Memory reuse across questions. Multiturn document VQA over successive user questions must combine document content with interaction history to resolve cross-turn references, carry forward comparison conditions, incorporate user corrections, and reuse evidence that remains applicable. MemoryDocDataSet explores the combination of conversational memory and reasoning over long text documents (Xie et al., 2026a). For visual documents, future systems could link dialogue to source pages and regions, retaining provenance, versions, and verified results to reuse earlier localization and reading while acquiring missing support for new questions. For example, a follow-up about another metric in the same table can reuse the table's location and measurement conventions. The key is to assess whether historical information remains applicable and revise affected memories after user corrections or document updates, reducing repeated processing and error propagation.

Continual, experience-driven strategy improvement. Beyond reusing specific evidence, systems could accumulate transferable processing strategies from experience across tasks. Because document types differ in structure and evidence dependencies, systems need to learn which retrieval, reading, and verification procedures suit which conditions. MemSkill revises memory skills through difficult cases (Zhang et al., 2026b), while AIDE^2 enables updated agents to continue participating in their own improvement (Srikanth et al., 2026), offering insights from general-purpose agents. Inspired by recursive self-improvement (RSI), future systems could propose strategies from document-QA successes, failures, and user feedback, validate them on independent tasks before updating a skill library or control procedure, and let the updated system participate in subsequent strategy revision. For example, experience with financial reports could yield a procedure that checks units and reporting periods before comparing values. The central challenge is to establish applicability conditions and feedback grounds that support strategy transfer to subsequent documents.

Table 4: Complete benchmark catalog: 55 records covering dataset size, question types, evidence annotations, and evaluation capabilities. Ordering and first-release dates follow Table 1. – denotes unreported or inapplicable information.

Benchmark			Data Size				Question Type				Evidence and Evaluation		
Name	**Venue**	**Domain**	**#Docs**	**#Pages**	**#Tokens**	**#QA**	**MD**	**MP**	**SP**	**Unans.**	**Source**	**Gran.**	**Eval.**
Retrieval-focused benchmarks													
Wiki-SS (2024a) *(24.06)*	EMNLP'24	Open	–	1.27M	–	52.7k	✗	✗	✓	✗	TXT	Page	Ans. Ret.
SlideVQA-open (2024a) *(24.06)*	EMNLP'24	Open	–	50.7k	–	12.4k	✗	✓	✓	✗	TXT/L/C/TAB/I	Page	Ret.
ViDoRe (2025) *(24.06)*	ICLR'25	Open	–	1.0k	–	3.81k	✗	✗	✓	✗	TXT/L/C/TAB/I	Page	Ans.[a] Ret.
REAL-MM-RAG (2025) *(25.02)*	ACL'25	Open	163	2.1k	–	4.6k	✗	✗	✓	✗	TXT/C/TAB/I	Page	Ans. Ret.
ViDoRe V2 (2025) *(25.03)*	arXiv'25	Open	62	1.1k	–	327	✓	✓	✓	✗	TXT/L/C/TAB/I	Page	Ret.
MMDocIR (2025) *(25.01)*	EMNLP'25	Closed	313	65.1	–	1.66k	✗	✓(18.9%)	✓(81.1%)	✗	TXT/L/C/TAB/I	Page, Elem.	Ret. Loc.
DocMMIR (2025) *(25.05)*	EMNLP'25[F]	Open	488.5k	–	537.4	488.5k	–	–	–	–	TXT/C/TAB/I	Doc.	Ret.[a]
ViMDoc (2026) *(25.10)*	ACL'26[F]	Open	1.4k	76.3k	–	10.9k	–	✓	✓	–	TXT/L/C/TAB/I	Page	Ret.
MultiDocR (2026b) *(26.05)*	KDD'26	Open	313	–	–	2.58k	–	✓(43.8%)	✓(56.2%)	✗	TXT/C/TAB/I	Page	Ans. Ret.
CMDR-Bench (2026) *(26.07)*	ECCV'26	Closed	255	183.5	–	800	✗	✓	–	✗	TXT/L/C/TAB/I	Page	Ret.
DocViRe (2026) *(26.08)*	arXiv'26	Open	8.29k	–	–	13.1k	–	–	–	–	TXT/C/TAB/I	Doc.	Ret.[a]
Answer-quality benchmarks: #Pages (avg.) = 1.0, #Pages (avg.) > 1.0 or unreported													
DocVQA (2021) *(20.07)*	WACV'21	Closed	12.8k	1.0	151.5	50k	✗	✗	✓	✗	TXT/L/C/TAB/I	–	Ans.
InfographicVQA (2022) *(21.04)*	WACV'22	Closed	5.5k	1.0	217.9	30k	✗	✗	✓	✗	TXT/L/C/TAB/I	–	Ans.
ChartQA (2022) *(22.03)*	ACL'22[F]	Closed	20.9k	1.0	–	32.7k	✗	✗	✓	✗	C	–	Ans.
ArXivQA* (2024) *(24.03)*	ACL'24	Closed	16.6k	1.0	–	100k	✗	✗	✓	✗	C/TAB/I	–	Ans.
DOGR-Bench (2025) *(24.11)*	ICCV'25	Closed	–	1.0	–	3.6k	✗	✗	✓	✗	TXT/L/C/I	Elem.	Ans. Loc.
GroundingDocQA-Bench (2026) *(26.06)*	CVPR'26	Closed	2.5k	1.0	–	5.0k	✗	✗	✓	✗	TXT/L/C/TAB/I	Elem.	Ans. Loc.[a]
DoCoG-Bench (2026) *(26.09)*	ECCV'26	Closed	3.0k	1.0	–	10.0k	✗	✗	✓	✗	TXT/L/C/TAB/I	Elem.	Ans. Loc.[a]
TAT-DQA (2022) *(22.07)*	MM'22	Closed	2.8k	1.1	–	16.6k	✗	✓	✓	✗	TXT/L/TAB	–	Ans.
SlideVQA (2023) *(23.01)*	AAAI'23	Closed	2,619	20.0	1.5k	14.5k	✗	✓(13.9%)	✓(86.1%)	✗	TXT/L/C/TAB/I	Page	Ans. Ret.
MP-DocVQA (2023) *(22.12)*	PR'23	Closed	5,928	8.3	2.0k	46.2k	✗	✗	✓	✗	TXT/L/C/TAB/I	Page	Ans. Ret.
DUDE (2023) *(23.05)*	ICCV'23	Closed	5,019	5.72	1.8k	41.5k	✗	✓(2.1%)	✓(85.2%)	✓(12.7%)	TXT/L/C/TAB/I	–	Ans.
UDA (2024) *(24.06)*	NIPS'24[D]	Closed	2.97k	–	–	29.6k	✗	–	✓	–	TXT/TAB	Elem.	Ans. Ret.[a]
MMLongBench-Doc (2024b) *(24.07)*	NIPS'24[D]	Closed	135	47.5	21.2k	1.1k	✗	✓(33.7%)	✓(45.7%)	✓(20.6%)	TXT/L/C/TAB/I	Page	Ans. Ret.
SPIQA* (2024) *(24.07)*	NIPS'24[D]	Closed	25.5k	–	–	266k	✗	–	–	✗	TXT/C/TAB/I	Elem.	Ans. Ret.[a]
VisRAG (2025) *(24.10)*	ICLR'25	Open	–	4.5k	–	3.6k	✗	✓	✓	✗	TXT/L/C/TAB/I	Page	Ans. Ret.
VisR-Bench (2024) *(24.11)*	ICLR'25	Closed	226	13.7	–	0.5k	✗	✗	✓	✗	TXT/C/TAB/I	Page	Ans. Ret.
PaperPDF (2026b) *(24.10)*	IJCV'26	Closed	70k	23	10.2k	1.1M	✗	✓	✓	✗	TXT/C/TAB/I	Elem.	Ans. Ret.[a]
VisDoMBench (2025) *(24.12)*	NAACL'25	Open	1,277	128.7	–	2.3k	✗	✓	✓	✗	TXT/L/C/TAB/I	Page	Ans. Ret.
OpenDocVQA (2025) *(25.04)*	CVPR'25	Open	–	231.3k	–	43.5k	✓	✓	✓	✗	TXT/L/C/TAB/I	Page	Ans. Ret.
DocBench (2025) *(24.07)*	KNLP'25[W]	Closed	229	66	46.4k	1.1k	✗	✓	✓	✓(11.3%)	TXT/L/C/TAB/I	–	Ans.
Wiki-VISA; Paper-VISA (2025a) *(24.12)*	ACL'25	Open	–	–	–	90k; 102.2k	✗	✗	✓	✓	TXT/TAB/I	Page, Elem.	Ans. Ret. Loc.
LongDocURL (2025) *(24.12)*	ACL'25	Closed	396	85.6	43.6k	2.3k	✗	✓(52.9%)	✓(47.0%)	✓(0.1%)	TXT/L/C/TAB/I	Page	Ans. Ret.
AIRQA-REAL (2025) *(25.05)*	ACL'25	Open/Closed	6.8k	–	–	553	✓(27%)	–	–	✗	TXT/TAB/I	Doc.	Ans. Ret.[a]
MMDocRAG* (2026a) *(25.05)*	NIPS'25[D]	Closed	222	67.0	–	4.06k	✗	✓(52.0%)	✓(48.0%)	✗	TXT/C/TAB/I	Page, Elem.	Ans. Ret.[a]
M-LongDoc (2025) *(24.11)*	EMNLP'25	Closed	180	210.8	121.0k	0.9k	✗	✗	✓	✗	TXT/C/TAB/I	Page	Ans. Ret.
ViDoSeek (2025) *(25.02)*	EMNLP'25	Open	300	6.0k	–	1.14k	✗	✗	✓	✗	TXT/L/C/TAB/I	Page	Ans. Ret.
FinRAGBench-V (2025) *(25.05)*	EMNLP'25	Open	2.21k	57.1k	–	1.39k	–	✓	✓	✗	TXT/C/TAB	Page, Elem.	Ans. Ret. Loc.
M3DocVQA (2025) *(24.11)*	ICCV'25[W]	Open	3,368	41.0k	–	2.4k	✓	✓	✓	✗	TXT/TAB/I	Doc.	Ans. Ret.[a]
Double-Bench (2026a) *(25.08)*	AAAI'26	Open	3.3k	72.9k	–	5.2k	–	✓	✓	✗	TXT/L/C/TAB/I	Page	Ans. Ret.
AirQA (2026) *(25.09)*	ICLR'26	Open/Closed	14.0k	–	–	1.2k	✓	–	–	✗	TXT/TAB/I	Doc.	Ans. Ret.[a]
M4DocBench (2026b) *(25.10)*	WWW'26	Open	303	306.4	–	158	✓(91.1%)	✓	✓	✗	TXT/C/TAB/I	Page, Elem.	Ans. Ret.[a] Loc.
DVisRAG (2026a) *(26.02)*	CVPR'26	Open	–	–	–	367.6k	✗	✓	✓	✗	TXT/L/C/TAB/I	Page	Ans. Ret. Rob.
VinQA (2026) *(26.06)*	CVPR'26	Open	–	5.0	–	44.3k	✓	✓	✓	✓	TXT/C/TAB/I	Page, Elem.	Ans. Ret. Loc.
ViDoRe V3 (2026) *(26.01)*	ACL'26	Open	–	–	–	3.1k	✓	✓	✓	–	TXT/L/C/TAB/I	Page, Elem.	Ans. Ret. Loc.
PaperScope* (2026c) *(26.04)*	ACL'26[F]	Open	2.2k	–	–	2.4k	✓(91.7%)	✓	–	✗	TXT/L/C/TAB/I	Doc.	Ans. Ret.[a]
M5BookVQA (2026) *(26.07)*	ACL'26	Open	–	16.8k	–	2.1k	–	✓	–	–	TXT/C/TAB/I	Page	Ans. Ret.
Wiki-CoE (2026b) *(26.05)*	SIGIR'26	Open	76.0k	5.0	–	70.4k	✓	✓	✗	✗	TXT/TAB/I/L	Page, Elem.	Ans. Ret. Loc.
MMLBD-C (2026) *(26.02)*	arXiv'26	Closed	135	47.5	21.2k	1.1k	✗	✓	✓	✓	TXT/L/C/TAB/I	Page	Ans. Ret.
MMDA (2026a) *(26.02)*	PACMMOD'26	Closed	537	–	–	1.1k	✗	–	–	–	TXT/L/C/TAB/I	–	Ans.
SynthDocBench (2026) *(26.07)*	COLM'26	Closed	200	51.1	–	1.8k	✗	✓	✓	✗	TXT/C/TAB	Elem.	Ans. Ret.[a]
DocVQA-2026 (2026) *(26.08)*	ICDAR'26[C]	Closed	–	–	–	240	✗	✓	✓	✓	TXT/L/C/TAB/I	–	Ans.
LFDocQA (2026c) *(26.04)*	arXiv'26	Open	964	–	–	1.0k	✗	✗	✓	✗	TXT/L/C/TAB	Page, Elem.	Ans. Ret.[a] Loc.
DocScope (2026) *(26.05)*	NIPS'26[D]	Closed	273	51.3	24.6k	1.12k	✗	✓(57.7%)	✓(35.3%)	✓(6.9%)	TXT/C/TAB/I/L	Page, Elem.	Ans. Ret. Loc.
XL-DocBench (2026a) *(26.07)*	NIPS'26	Open	331	–	227.5k	1.5k	✓(10.9%)	✓(72.6%)	✓	✓(14.4%)	TXT/C/TAB/I	Page	Ans. Ret.

1. Groups and dates. Benchmarks are grouped by primary evaluation target and input scale. Ordering follows Table 1. (YY.MM) denotes the first-release month, preferentially the first arXiv version; an earlier official release, or the official release when no preprint exists, takes precedence.

2. Venue. Superscripts F/W/D/S/I/C denote Findings, Workshop, Datasets and Benchmarks, Short, Industry, and competition reports, respectively.

3. Scale and input. Open and Closed denote a cross-document candidate collection and a given single document; Open/Closed supports both. #Docs counts documents or independent page inputs. #Pages is the average number of input pages per question, using a document-level mean when per-question statistics are unavailable; Open counts the full candidate collection for that question, and an independent image counts as one page. #Tokens is the mean text-token count per input document. #QA counts questions, queries, or localization tasks.

4. Question Type. MD, MP, and SP denote multi-document, multi-page, and single-page evidence requirements, respectively; Unans. denotes questions unanswerable from the given evidence. ✓/✗ indicates inclusion/exclusion; parentheses give proportions. In Source, TXT/L/C/TAB/I denotes text/layout/charts/tables/images. Gran. denotes the verification granularity supported by public evidence annotations. In open-domain settings, page annotations with document identities also support document verification, which is not listed again.

5. Evaluation. Eval. abbreviates Evaluation. Capabilities supported by the benchmark and its public annotations: Ans., answer quality; Ret., evidence retrieval, typically page/whole-image hits, recall, or ranking; Loc., within-page element or region localization, such as IoU-based evaluation; Eff., processing efficiency and cost; Rob., robustness to controlled perturbations. **Exceptions (a):** ViDoRe supports Ans. only on subsets with public answers. For Ret., SPIQA retrieves figures/tables; DocMMIR, DocViRe, PaperScope, AIRQA-REAL, and AirQA retrieve documents; M3DocVQA measures recall by the documents containing retrieved pages; M4DocBench covers documents, pages, and layout elements; SynthDocBench matches chart/element IDs; UDA measures text-evidence coverage; MMDocRAG includes text/image-segment selection; LFDocQA includes element-ID matching; PaperPDF matches paragraph/figure/table IDs. Loc. for GroundingDocQA-Bench and DoCoG-Bench includes segmentation masks.

6. Pre-extracted inputs (*). ArXivQA uses pre-extracted scientific figures; SPIQA uses pre-extracted figure/table images and text; MMDocRAG generation evaluation uses pre-extracted text/image segments; PaperScope supplies pre-parsed paper text and images. These settings do not directly evaluate the complete parsing process from raw pages.

Table 5: Complete comparison of 29 methods without explicit retrieval, grouped by their primary single-page or multipage setting. Ordering and first-release dates follow Table 1. + denotes combinations and / alternatives; – denotes unconfirmed information.

Method		Parsing		Generation & Learning		Eval.
Name	**Venue**	**OCR**	**Deep**	**Agent Model**	**Learning**	
Single-page VQA						
VisionTaPas (2022) *(22.03)*	ACL'22[F]	✓	✗	Single TAPAS	SFT	Ans.
Donut (2022) *(21.11)*	ECCV'22	✗	✗	Single BART	Pretraining + SFT	Ans. Eff.
LayoutLMv3 (2022) *(22.04)*	MM'22	✓	✗	Single RoBERTa	Pretraining + SFT	Ans. Loc.[a]
UDOP (2023) *(22.12)*	CVPR'23	✓	✗	Single T5	Pretraining + SFT	Ans.
Pix2Struct (2023) *(22.10)*	ICML'23	✗	✗	Single Pix2Struct	Pretraining + SFT	Ans.
UReader (2023) *(23.10)*	EMNLP'23[F]	✗	✗	Single Llama	SFT	Ans. Eff.
LayoutLLM (Fujitake) (2024) *(24.03)*	LREC-COLING'24	✓	✗	Single Llama	SFT	Ans.
mPLUG-DocOwl 1.5 (2024) *(24.03)*	EMNLP'24[F]	✗	✗	Single Llama	Pretraining + SFT	Ans. Eff.
DocLayLLM (2025) *(24.08)*	CVPR'25	✓	✗	Single Llama	Pretraining + SFT	Ans. Eff.
DocVLM* (2025) *(24.12)*	CVPR'25	✓	✗	Single Qwen-VL / InternVL / LLaVA	SFT	Ans. Eff.
DOGR (2025) *(24.11)*	ICCV'25	✗	✗	Single Qwen	Pretraining + SFT	Ans. Loc.
M3Grounder (2026) *(26.06)*	CVPR'26	✗	✗	Single InternVL / Qwen-VL	SFT	Ans. Loc.[a]
DocVAL (2025) *(25.11)*	ICML'26	✗	✗	Single Gemma3	SFT	Ans. Loc. Eff.
Uni-DocRobust (2026b) *(26.--)*	ICML'26	✗	✗	Single LLaVA / Qwen-VL / InternVL	Pretraining	Ans. Eff. Rob.
DoCoG (2026) *(26.09)*	ECCV'26	✗	✗	Single Qwen-VL	SFT + RL	Ans. Loc.[a]
RegionDoc-R1* (2026a) *(26.--)*	EMNLP'26	✗	✗	Single Qwen-VL	SFT + RL	Ans.
Multi-page VQA						
M3D (2023) *(23.01)*	AAAI'23	✓	✗	Single T5	SFT	Ans. Ret.
Hi-VT5 (2023) *(22.12)*	PR'23	✓	✗	Single T5	Pretraining + SFT	Ans. Ret.
GRAM (2024) *(24.01)*	CVPR'24	✓	✗	Single T5	SFT	Ans. Eff.
Docopilot (2025) *(25.06)*	CVPR'25	✗	✗	Single InternVL	SFT	Ans. Eff.
mPLUG-DocOwl2 (2025) *(24.09)*	ACL'25	✗	✗	Single Llama	Pretraining + SFT	Ans. Eff.
DocR1 (2026b) *(25.08)*	AAAI'26	✗	✗	Single Qwen-VL	RL	Ans. Ret.
LAT (2026c) *(25.11)*	AAAI'26	✗	✗	Single Qwen-VL	SFT + RL	Ans. Ret. Loc. Eff.
DocSLM (2026) *(25.11)*	CVPR'26[F]	✓	✗	Single Qwen	Pretraining + SFT	Ans. Eff.
DocSeeker (2026b) *(26.04)*	CVPR'26	✗	✗	Single Qwen-VL	SFT + RL	Ans. Ret. Eff. Rob.
VinQA (Page Encoding) (2026) *(26.06)*	CVPR'26	✗	✗	Single Qwen-VL	SFT	Ans. Ret.[a] Eff.
CoE (2026b) *(26.05)*	SIGIR'26	✗	✗	Single Qwen-VL	SFT	Ans. Ret. Loc.[a] Eff.
OriOn (2026) *(26.02)*	arXiv'26	✗	✗	Single Mistral / Qwen-VL	Pretraining + SFT + RL	Ans. Eff.
Quote-and-Retrieve (2026e) *(26.07)*	arXiv'26	✗	✗	Single Qwen-VL	RL	Ans. Ret. Loc. Eff.

1. Methods. * marks methods primarily evaluated on single pages/images that also support multiple pages. Venue superscripts follow Table 4; (YY.MM) denotes first release, and – in the month position means unavailable. Deep = Deep Parsing; Eval. = Evaluation.

2. Model and learning. Model gives the main answer model or language-backbone family. Learning records added Pretraining, supervised fine-tuning (SFT), or reinforcement learning (RL).

3. Evaluation. Ans./Ret./Loc./Eff./Rob. denote answer quality/retrieval/localization/efficiency/robustness. Only dimensions quantitatively reported in the original paper are listed. Ret. denotes page-level evidence selection or prediction and does not imply explicit retrieval in the answering pipeline.
Exceptions (a): LayoutLMv3 Loc. is layout-box detection; VinQA (Page Encoding) Ret. is element-citation F1; CoE Loc. includes box-center hits; M3Grounder and DoCoG Loc. include segmentation masks.

Table 6: Complete comparison of methods with explicit retrieval: 20 retrieval-only and 90 document VQA methods. Ordering and date conventions follow Table 1; venue superscripts follow Table 4. Mechanism specifies retrievers, evidence mappings, and retrieval training; Model gives the main generator model family. + denotes combinations and / alternatives; – denotes unreported, unconfirmed, or inapplicable information.

Method		Parsing		Evidence Organization and Retrieval				Generation & Learning				Eval.
Name	Venue	OCR	Deep	Modality	Gran.	Mechanism	Acq.	Agent	Reasoning	Model	Learning	
Retrieval-only methods												
DSE (2024a) *(24.06)*	EMNLP'24	✗	✗	Uni(V)	Page	$DSE^{t}(P^{V})$	One-shot	–	–	–	–	Ret. Eff.
ColPali (2025) *(24.06)*	ICLR'25	✗	✗	Uni(V)	Page	$ColPali^{t}(P^{V})$	One-shot	–	–	–	–	Ret. Eff.
RobTabCol (2025) *(25.02)*	ACL'25	✗	✗	Uni(V)	Page	$RobTabCol^{t}(P^{V})$	One-shot	–	–	–	–	Ret. Rob.
Light-ColPali (2025b) *(25.06)*	$ACL'25^{F}$	✗	✗	Uni(V)	Page	$ColPali^{t}(P^{V})$	One-shot	–	–	–	–	Ret. Eff.
DocMMIR (2025) *(25.05)*	$EMNLP'25^{F}$	✓	✗	Mixed(T/V)	Doc.	$OpenCLIP^{t}(D^{TV})$	One-shot	–	–	–	–	Ret.[a] Eff.
Evo-Retriever (2026c) *(26.03)*	CVPR'26	✗	✗	Uni(V)	Page	$Evo\text{-}Retriever^{t}(P^{V})$	One-shot	–	–	–	–	Ret.
HEAVEN (2026) *(25.10)*	$ACL'26^{F}$	✗	Summary	Uni(V)	Page	$DSE(P^{V}) + ColQwen(P^{V})$	One-shot	–	–	–	–	Ret.[a] Eff.
ReAlign (2026a) *(26.04)*	SIGIR'26	✗	✗	Uni(V)	Page	$ReAlign^{t}(P^{V})$	One-shot	–	–	–	–	Ret. Loc.[a]
ZipRerank (2026) *(26.05)*	ICML'26	✗	✗	Uni(V)	Page	$DSE(P^{V}) + ZipRerank^{t}(P^{V})$	One-shot	–	–	–	–	Ret. Eff.
DocRetriever (2026b) *(26.05)*	KDD'26	✗	✗	Uni(V)	Page	$ColQwen(P^{V}) + VLM(P^{V})$	One-shot	–	–	–	–	Ret. Eff.
LMS-Retrieval (2026) *(26.08)*	ICDAR'26	✓	Tree	Uni(T)	Elem.	$LMS\text{-}Retrieval^{t}(E^{T})$	One-shot	–	–	–	–	Ret.[a] Loc.
LightSTAR (2026) *(26.06)*	ECCV'26	✗	✗	Uni(V)	Page	$LightSTAR^{t}(P^{V})$	One-shot	–	–	–	–	Ret. Eff.
CMDR-Embed (2026) *(26.07)*	ECCV'26	✗	✗	Uni(V)	Page	$CMDR\text{-}Embed^{t}(P^{V})$	One-shot	–	–	–	–	Ret. Eff.
AdaMerge (2026) *(26.09)*	CIKM'26	✗	✗	Uni(V)	Page	$ColQwen/ColNomic(P^{V})$	One-shot	–	–	–	–	Ret. Eff.
MMRetHeads (2026a) *(26.05)*	EMNLP'26	✗	✗	Uni(V)	Page, Elem.	$MMRetHeads(P^{V}; E^{V})$	One-shot	–	–	–	–	Ret. Loc.[a]
MIDR (2026) *(26.09)*	EMNLP'26	✓	Summary	Uni(T)	Page	$BM25F(P^{T}) + EmbeddingGemma(P^{T})$	One-shot	–	–	–	–	Ret. Eff.
DocPC (2026) *(26.08)*	arXiv'26	✗	✗	Uni(V)	Doc.	$ColQwen^{t}(D^{V})$	One-shot	–	–	–	–	Ret.[a] Eff.
MarginMerge (2026) *(26.08)*	arXiv'26	✗	✗	Uni(V)	Page	$MarginMerge^{t}(P^{V})$	One-shot	–	–	–	–	Ret. Eff.
DistilVDR (2026d) *(26.08)*	$EMNLP'26^{W}$	✗	✗	Uni(V)	Page	$DistilVDR^{t}(P^{V})$	One-shot	–	–	–	–	Ret. Eff.
ConceptFormer (2026) *(26.08)*	arXiv'26	✗	✗	Uni(V)	Page	$ConceptFormer^{t}(P^{V})$	One-shot	–	–	–	–	Ret.
Document VQA methods												
MHST (2022) *(22.07)*	MM'22	✓	✗	Uni(T)	Page, Elem.	$TF\text{-}IDF(E^{T})$	One-shot	Single	Single-pass	LayoutLM	SFT	Ans.
CREAM (2024) *(24.10)*	ACM MM'24	✓	✗	Uni(T)	Page, Elem.	$BGE(E^{T}\rightarrow[E^{T}, P^{V}]) + RankVicuna(E^{T}\rightarrow[E^{T}, P^{V}])$	One-shot	Single	Single-pass	Llama	SFT	Ans.
VisRAG (2025) *(24.10)*	ICLR'25	✗	✗	Uni(V)	Page	$VisRAG\text{-}Ret^{t}(P^{V})$	One-shot	Single	Single-pass	MiniCPM-V/GPT-4o	✗	Ans. Ret. Eff.
SV-RAG (2024) *(24.11)*	ICLR'25	✗	✗	Uni(V)	Page	$Col\text{-}Ret^{t}(P^{V})$	One-shot	Single	Single-pass	PaliGemma/ Phi-Vision/ InternVL	SFT	Ans. Ret. Eff.
PDF-WuKong (2026b) *(24.10)*	IJCV'26	✓	✗	Mixed(T/V)	Elem.	$PDF\text{-}WuKong^{t}(E^{TV})$	One-shot	Single	Single-pass	InternLM-XComposer	SFT	Ans. Eff.
VisDoMRAG (2025) *(24.12)*	NAACL'25	✓	✗	Mixed(T/V)	Page, Elem.	$BGE(E^{T}) + ColQwen(P^{V})$	One-shot	Multi	Single-pass	Qwen-VL/ Gemini/ GPT-4o	✗	Ans. Ret.
VDocRAG (2025) *(25.04)*	CVPR'25	✗	✗	Uni(V)	Page	$VDoc\text{-}Ret^{t}(P^{V})$	One-shot	Single	Single-pass	Phi-Vision	SFT	Ans. Ret. Eff.
SuperRAG (2025) *(25.02)*	$NAACL'25^{I}$	✓	Graph, Tree	Uni(T)	Page, Elem.	$OpenAI\ Embedding(P^{T}; E^{T}) + BM25(E^{T}) + Cohere(E^{T}) + LLM(P^{T}; E^{T})$	One-shot	Single	Single-pass	GPT-4o	✗	Ans.
Doc-React (2025a) *(24.12)*	$ACL'25^{S}$	✗	✗	Uni(V)	Page	$ColPali(P^{V}) + VLM(P^{V})$	Iterative	Single	Multi-step	GPT-4o	✗	Ans.
VISA (2025a) *(24.12)*	ACL'25	✗	✗	Uni(V)	Page	$DSE(P^{V})$	One-shot	Single	Single-pass	Qwen-VL	SFT	Ans. Loc.
NeuSym-RAG (2025) *(25.05)*	ACL'25	✓	Graph, Summary	Mixed(T/V)	Doc., Page, Elem.	$BM25(P^{T}; E^{T}) + MiniLM(P^{T}; E^{T}) + BGE(P^{T}; E^{T}) + CLIP(E^{V}) + VLM(D^{T}; P^{TV}; E^{TV})$	Iterative	Single	Multi-step	Qwen-VL	✗	Ans. Ret.[a] Eff.
MDocAgent (2025) *(25.03)*	arXiv'25	✓	✗	Mixed(T/V)	Page	$ColBERT(E^{T}\rightarrow P^{T}) + ColPali(P^{V})$	One-shot	Multi	Single-pass	Llama + Qwen-VL	✗	Ans.
FRAG (2025) *(25.04)*	arXiv'25	✗	✗	Uni(V)	Page	$VLM(P^{V})$	One-shot	Single	Single-pass	LLaVA/ InternVL	✗	Ans.
VRAG-RL (2026d) *(25.05)*	NIPS'25	✗	✗	Uni(V)	Page, Elem.	$ColPali(P^{V}) + VLM^{t}(E^{V})$	Iterative	Single	Multi-step	Qwen-VL	SFT + RL	Ans. Ret. Eff.
Dual-Stream RAG (2025) *(26.01)*	APWeb-WAIM'25	✓	✗	Mixed(T/V)	Page	$NV\text{-}Embed(E^{T}\rightarrow P^{V}) + ColQwen(P^{V}) + VLM(P^{V})$	One-shot	Multi	Single-pass	Qwen-VL	✗	Ans. Ret.
Retrieval-aware Tuning (2025) *(24.11)*	EMNLP'25	✓	✗	Uni(V)	Page, Elem.	$ColPali(P^{V}\rightarrow[P^{T}, E^{V}])$	One-shot	Single	Single-pass	Qwen-VL	SFT	Ans. Ret. Rob.
ViDoRAG (2025) *(25.02)*	EMNLP'25	✓	✗	Mixed(T/V)	Page	$NV\text{-}Embed(E^{T}\rightarrow P^{V}) + ColQwen(P^{V}) + VLM(P^{V})$	One-shot	Multi	Multi-step	GPT-4o/ Qwen-VL/ Llama-Vision	✗	Ans. Ret. Eff.
RGenCite (2025) *(25.05)*	EMNLP'25	✗	✗	Uni(V)	Page	$ColQwen(P^{V})$	One-shot	Single	Single-pass	Qwen-VL	✗	Ans. Ret. Loc.[a]
SimpleDoc (2025) *(25.06)*	EMNLP'25	✓	Summary	Mixed(T/V)	Page	$ColQwen(P^{V}\rightarrow P^{TV}) + LLM(P^{T}\rightarrow P^{TV})$	Iterative	Single	Multi-step	Qwen-VL	✗	Ans. Ret. Eff.
MoLoRAG (2025b) *(25.09)*	EMNLP'25	✗	Graph	Uni(V)	Page	$ColPali(P^{V}) + VLM(P^{V})$	One-shot	Single	Single-pass	Qwen-VL/ DeepSeek-VL/LLaVA	✗	Ans. Ret. Eff.
LILaC (2025) *(25.11)*	EMNLP'25	✓	Graph	Mixed(T/V)	Elem.	$MM\text{-}Embed(E^{TV}; E^{V})$	One-shot	Single	Single-pass	Qwen-VL	✗	Ans. Ret.[a] Eff.
DocAgent (2025a) *(25.11)*	EMNLP'25	✓	Tree	Uni(T)	Page, Elem.	$Keyword(E^{T}) + VLM(E^{T}\rightarrow[E^{TV}, P^{V}])$	Iterative	Multi	Multi-step	GPT-4o/ Claude/ Gemini	✗	Ans. Ret. Eff. Rob.
M3DocRAG (2025) *(24.11)*	$ICCV'25^{W}$	✗	✗	Uni(V)	Page	$ColPali(P^{V})$	One-shot	Single	Single-pass	Qwen-VL	✗	Ans. Ret. Eff.
MMGraphRAG (2025) *(25.07)*	arXiv'25	✓	Graph	Uni(T)	Elem.	$–(E^{T}\rightarrow E^{TV})$	One-shot	Single	Single-pass	Qwen + InternVL	✗	Ans. Eff.
RegionRAG (2026e) *(25.10)*	AAAI'26	✗	✗	Uni(V)	Elem.	$RegionRetriever^{t}(E^{V})$	One-shot	Single	Single-pass	Qwen-VL	✗	Ans. Ret. Eff.
URaG (2026) *(25.11)*	AAAI'26	✗	✗	Uni(V)	Page	$URaG^{t}(P^{V})$	One-shot	Single	Single-pass	Qwen-VL	SFT	Ans. Ret. Eff.

Continued on next page

Table 6 (continued)

Method		Parsing		Evidence Organization and Retrieval				Generation & Learning				Eval.
Name	**Venue**	**OCR**	**Deep**	**Modality**	**Gran.**	**Mechanism**	**Acq.**	**Agent**	**Reasoning**	**Model**	**Learning**	
M3RAG (2026) *(26.02)*	MMM'26	✗	✗	Mixed(T/V)	Page, Elem.	–(P^V; E^{TV})	Iterative	Multi	Multi-step	Mistral + (MiniCPM-V/ Qwen-VL)	✗	Ans. Eff.
Agentic Hybrid (EXTRACTOR) (2026) *(25.09)*	ICLR'26	✓	Graph	Uni(T)	Doc., Page, Elem.	MiniLM(E^T) + LLM^t(D^T; P^T; E^T)	Iterative	Single	Multi-step	Qwen	SFT	Ans. Ret.[a] Eff.
SCAN (2026) *(25.05)*	EACL'26[F]	✗	✗	Uni(V)	Elem.	ColQwen(E^V)	One-shot	Single	Single-pass	Qwen-VL	✗	Ans. Loc.
SCoPE VLM (2026) *(25.10)*	EACL'26	✗	✗	Uni(V)	Page	VLM^t(P^V)	Iterative	Single	Multi-step	Qwen-VL	SFT + RL	Ans. Eff.
Doc-Researcher (2026b) *(25.10)*	WWW'26	✓	Summary	Mixed(T/V)	Doc., Page, Elem.	Jina(P^T; E^T; E^V→E^T) + LLM(D^T)	Iterative	Multi	Multi-step	Qwen/ DeepSeek-R1	✗	Ans. Ret.[a] Loc. Eff.
DMAP (2026) *(26.01)*	WWW'26	✓	Tree, Summary	Mixed(T/V)	Page, Elem.	ColBERT(E^T→P^T) + ColPali(P^V) + LLM(P^T→P^{TV}; E^T→E^{TV})	Iterative	Multi	Multi-step	GPT-4o/ Qwen	✗	Ans.
RAG-Anything (2025) *(25.10)*	arXiv'25	✓	Graph	Uni(T)	Elem.	OpenAI Embedding(E^T→E^{TV}) + BGE-reranker(E^T→E^{TV})	One-shot	Single	Single-pass	GPT-4o	✗	Ans.
MACT (2026) *(25.08)*	CVPR'26	–	✗	–	–	Search(D^-, E^-); Vector/ Keyword(E^T)	–	Multi	Multi-step	Qwen-VL + Qwen/ MiMo-VL + MiMo/ InternVL	SFT + RL	Ans. Eff.
SLEUTH (2026a) *(25.11)*	CVPR'26	✗	✗	Uni(V)	Page, Elem.	ColPali(P^V) + VLM(P^V→[P^V, E^T])	One-shot	Multi	Single-pass	Qwen-VL/ GLM-V/ Gemini	✗	Ans.
RobustVisRAG (2026a) *(26.02)*	CVPR'26	✗	✗	Uni(V)	Page	RobustVisRAG^t(P^V)	One-shot	Single	Single-pass	MiniCPM-V	✓	Ans. Ret. Rob.
DocPrune (2026) *(26.04)*	CVPR'26	✗	✗	Uni(V)	Page	ColPali(P^V)	One-shot	Single	Single-pass	Qwen-VL	✗	Ans. Eff.
HKRAG (2025) *(25.11)*	arXiv'25	✗	✗	Uni(V)	Page	HKRAG-Ret^t(P^V) + VLM(P^V)	One-shot	Multi	Multi-step	Phi-Vision	✗	Ans. Ret. Eff.
CogDoc (2025) *(25.12)*	arXiv'25	✗	✗	Uni(V)	Page	VLM^t(P^V)	One-shot	Single	Single-pass	Qwen-VL	RL	Ans. Ret.
LAD-RAG (2026) *(25.10)*	ACL'26	✓	Graph	Uni(T)	–	Vector(E^T→–) + LLM (E^T→–)	Iterative	Single	Multi-step	GPT-4o/ Phi-Vision/ Pixtral/ InternVL	✗	Ans. Ret. Eff.
ALDEN (2026b) *(25.10)*	ACL'26	✗	✗	Uni(V)	Page	ColQwen(P^V) + VLM^t(P^V)	Iterative	Single	Multi-step	Qwen-VL	RL	Ans. Ret. Eff.
SlideAgent (2026) *(25.10)*	ACL'26	✓	Tree, Summary	Uni(T)	Page, Elem.	SFR(P^T→P^{TV}; E^T→[P^V, E^T])	One-shot	Multi	Single-pass	GPT-4o/ InternVL	✗	Ans. Ret. Eff.
DocLens (2026a) *(25.11)*	ACL'26	✓	✗	Mixed(T/V)	Page, Elem.	VLM(P^{TV}→[P^{TV}, E^V])	One-shot	Multi	Single-pass	Gemini/ Claude	✗	Ans. Ret. Loc.
Lang2Act (2026d) *(26.01)*	ACL'26[F]	✗	✗	Uni(V)	Page	ColQwen(P^V)	One-shot	Single	Single-pass	Qwen-VL	RL	Ans. Ret. Loc.[a] Eff.
MAB-DQA (2026c) *(26.04)*	ACL'26	✗	Graph	Uni(V)	Page	ColPali(P^V) + VLM(P^V)	Iterative	Single	Multi-step	Qwen-VL	✗	Ans. Ret. Eff.
Doc-V* (2026) *(26.04)*	ACL'26	✗	✗	Uni(V)	Page	ColQwen(P^V) + VLM^t(P^V)	Iterative	Single	Multi-step	Qwen-VL	SFT + RL	Ans. Ret. Eff.
MM-Doc-R1 (2026) *(26.04)*	ACL'26[F]	✓	Tree	Uni(T)	Page	BM25(P^T→P^{TV})	Iterative	Multi	Multi-step	Qwen + Qwen-VL	RL	Ans. Ret. Eff.
HiKEY (2026) *(26.05)*	ACL'26	✓	Tree	Mixed(T/V)	Elem.	BM25(E^T) + GTE(E^T) + MM-Embed(E^V)	One-shot	Single	Single-pass	Qwen-VL	✗	Ans. Ret.[a] Eff.
TRACE (2026) *(26.07)*	ACL'26	✗	Graph	Uni(V)	Page	ColPali(P^V) + VLM(P^V)	One-shot	Multi	Single-pass	Qwen-VL	✗	Ans. Ret. Eff.
CAPS (2026b) *(26.07)*	ACL'26	✗	✗	Uni(V)	Page	VLM^t(P^V)	One-shot	Single	Single-pass	Qwen-VL/ LLaVA/ DeepSeek-VL	✗	Ans. Ret. Eff.
FUMA (2026b) *(26.07)*	ACL'26	✗	✗	Uni(V)	Page	FUMA^t(P^V)	One-shot	Single	Single-pass	Qwen	SFT	Ans. Ret. Eff.
MDocRAG-RL (2026) *(26.07)*	ACL'26[F]	✗	✗	Uni(V)	Page, Elem.	MDocRAG-RL^t(P^V) + VLM^t(E^V)	Iterative	Single	Multi-step	Qwen-VL	SFT + RL	Ans. Ret. Eff.
G^2-Reader (2026) *(26.01)*	ICML'26	✓	Graph	Uni(T)	Elem.	Dense(E^T→E^{TV})	Iterative	Multi	Multi-step	Qwen-VL	✗	Ans. Ret.[a] Eff.
VimRAG (2026e) *(26.02)*	ICML'26	✗	✗	Uni(V)	Page	GVE(P^V) + VLM^t(P^V)	Iterative	Single	Multi-step	Qwen-VL	SFT + RL	Ans. Ret. Eff.
UniDocVLM (2026) *(26.08)*	KDD'26	✗	✗	Uni(V)	Page	VisRAG-Ret(P^V)	One-shot	Single	Single-pass	Qwen-VL	SFT + RL	Ans.
MLDocRAG (2026) *(26.02)*	arXiv'26	✓	Graph	Uni(T)	Elem.	BGE(E^T→E^{TV})	One-shot	Single	Single-pass	Qwen-VL	✗	Ans.
MM-BizRAG (2026) *(26.06)*	ACL'26[I]	✓	✗	Uni(T)	Page, Elem.	OpenAI(E^T→E^{TV}; P^T→P^{TV}) + BM25(E^T→E^{TV}; P^T→P^{TV}) + LLM(E^T→E^{TV}; P^T→P^{TV})	One-shot	Single	Single-pass	GPT-4.1	✗	Ans. Eff.
MoDora (2026a) *(26.02)*	PACMMOD '26	✓	Tree, Summary	Mixed(T/V)	Elem.	Qwen-Embedding(E^T) + LLM(E^T→E^{TV}) + VLM (E^{TV})	One-shot	Single	Single-pass	GPT-5	✗	Ans. Ret.[a] Eff.
AdaNav (2026b) *(26.08)*	ICDAR'26	–	Tree	–	Page, Elem.	–	–	Multi	–	–	–	Ans.
VisRAG 2.0 (2025b) *(25.10)*	ACM MM'26	✗	✗	Uni(V)	Page	VisRAG-Ret(P^V)	One-shot	Single	Single-pass	Qwen-VL	SFT + RL	Ans.
VISOR (2026b) *(26.04)*	ACM MM'26	✗	✗	Uni(V)	Page, Elem.	ColQwen(P^V) + VLM^t(E^V)	Iterative	Single	Multi-step	Qwen-VL	SFT + RL	Ans. Ret. Eff.
LFRAG (2026c) *(26.04)*	arXiv'26	✗	Graph	Uni(V)	Elem.	LFRAG^t(E^V)	One-shot	Single	Single-pass	Qwen-VL	✗	Ans. Ret.[a] Eff.
BayesRAG (2026d) *(26.01)*	NIPS'26	✓	Graph	Mixed(T/V)	Page, Elem.	Qwen-Embedding(E^T) + BGE-reranker(E^T) + PE (E^V) + ColNomic(P^V)	One-shot	Single	Single-pass	GPT-4o	✗	Ans. Ret. Eff.
AgenticOCR (2026f) *(26.02)*	NIPS'26	✓	✗	Mixed(T/V)	Page, Elem.	Qwen-VL-Reranker(P^V) + VLM^t(E^V→E^{TV})	One-shot	Single	Multi-step	Qwen-VL + Gemini	SFT + RL	Ans. Ret. Loc. Eff.
DocAtlas (2026b) *(26.07)*	NIPS'26	✓	Tree, Summary	Mixed(T/V)	Page, Elem.	LLM(E^T→[P^{TV}, E^V]) + VLM^t(P^{TV}; E^V)	Iterative	Single	Multi-step	Qwen-VL	RL	Ans. Ret. Eff.
DocDancer (2026d) *(26.01)*	EMNLP'26	✓	Tree	Uni(T)	Page, Elem.	Keyword(E^T→[E^{TV}, P^V])	Iterative	Single	Multi-step	Qwen + Qwen-VL	SFT	Ans.

Continued on next page

Table 6 (continued)

Method		Parsing		Evidence Organization and Retrieval				Generation & Learning				Eval.
Name	**Venue**	**OCR**	**Deep**	**Modality**	**Gran.**	**Mechanism**	**Acq.**	**Agent**	**Reasoning**	**Model**	**Learning**	
Agent Consolidation (2026a) *(26.01)*	EMNLP'26[F]	✓	–	Uni(T)	Page, Elem.	Keyword(E^T) + LLM^t ($E^T \rightarrow [E^{TV}, P^V]$)	Iterative	Single	Multi-step	Qwen	SFT	Ans. Eff.
AWM (2026a) *(26.08)*	EMNLP'26[F]	✗	✗	Uni(V)	Page	Jina(P^V)	Iterative	Single	Multi-step	Qwen-VL	SFT + RL	Ans.
Doc-REFRAG (2026a) *(26.08)*	EMNLP'26	✗	✗	Uni(V)	Page	Jina(P^V)	One-shot	Single	Single-pass	Llama	SFT + RL	Ans. Ret. Eff.
Doc-Search (2026c) *(26.06)*	arXiv'26	✓	✗	Uni(V)	Page	ColPali($P^V \rightarrow P^T$)	Iterative	Single	Multi-step	Qwen	RL	Ans. Ret. Eff.
Hybrid Retriever Evolution (2026a) *(26.06)*	arXiv'26	✓	✗	Mixed(T/V)	Page, Elem.	BM25(E^T) + ColBERT(E^T) + ColPali(P^V) + VLM(P^V)	Iterative	Single	Multi-step	Gemini/ GPT-5	✗	Ans. Ret.
MARDoc (2026b) *(26.06)*	arXiv'26	✓	Tree	Mixed(T/V)	Page, Elem.	Keyword(E^T) + VLM(P^V; E^{TV})	Iterative	Multi	Multi-step	Qwen-VL	✗	Ans. Eff.
CDS (2026a) *(26.06)*	arXiv'26	✓	Graph	Uni(T)	Elem.	Nomic($E^T \rightarrow E^{TV}$)	One-shot	Single	Single-pass	Qwen-VL	✗	Ans.
FlowReader (2026b) *(26.06)*	arXiv'26	✓	Graph	Uni(T)	Elem.	Nomic($E^T \rightarrow E^{TV}$) + BM25 ($E^T \rightarrow E^{TV}$) + VLM($E^T \rightarrow E^{TV}$)	Iterative	Multi	Multi-step	Qwen-VL	✗	Ans. Eff.
EviProp (2026c) *(26.06)*	arXiv'26	✓	Graph	Mixed(T/V)	Page	ColPali(P^V; $E^V \rightarrow P^V$) + OpenAI Embedding($E^T \rightarrow P^V$)	One-shot	Single	Single-pass	Qwen-VL	✗	Ans. Ret. Eff.
MAGE-RAG (2026) *(26.06)*	arXiv'26	✓	Graph, Summary	Mixed(T/V)	Page, Elem.	ColPali(P^V) + VLM($P^T \rightarrow P^{TV}$; $E^T \rightarrow E^{TV}$)	Iterative	Single	Multi-step	Qwen-VL	✗	Ans. Eff.
Six-tool Agent (2026) *(26.09)*	EMNLP'26[I]	✓	Tree	Uni(T)	Page, Elem.	BM25(E^T) + VLM($E^T \rightarrow [P^V, E^{TV}]$)	Iterative	Single	Multi-step	Claude/ Qwen	✗	Ans. Eff.
TAP-RAG (2026) *(26.07)*	arXiv'26	✓	Graph	Uni(T)	Page, Elem.	text-embedding($E^T \rightarrow [E^T, P^V]$)	One-shot	Single	Single-pass	Qwen-VL	✗	Ans.
HIEVI-RAG (2026a) *(26.07)*	arXiv'26	✗	✗	Uni(V)	Page	ColQwen(P^V) + VLM^t(P^V)	One-shot	Single	Multi-step	Qwen-VL	RL	Ans. Ret.
HierDoc (2026) *(26.07)*	arXiv'26	✓	✗	Mixed(T/V)	Page, Elem.	VLM^t(P^V; E^{TV})	One-shot	Single	Single-pass	Qwen-VL	✗	Ans. Ret.[a] Eff.
DocTrace (2026a) *(26.08)*	arXiv'26	✓	✗	Uni(V)	Page, Elem.	VLM^t($P^V \rightarrow [P^V, E^T]$)	One-shot	Single	Single-pass	Qwen-VL	SFT + RL	Ans. Ret. Eff.
DocMemo (2026b) *(26.08)*	arXiv'26	✓	Tree, Summary	Mixed(T/V)	Page	ColQwen($P^V \rightarrow P^{TV}$) + LLM($P^T \rightarrow P^{TV}$)	Iterative	Multi	Multi-step	Qwen-VL	✗	Ans. Ret. Eff.
InSight-doc (2026b) *(26.08)*	arXiv'26	✗	✗	Uni(V)	Page, Elem.	VLM^t(E^V)	One-shot	Single	Multi-step	Qwen-VL	SFT + RL	Ans. Loc. Eff.
TRIDENT (2026a) *(26.08)*	arXiv'26	✓	Graph, Tree, Summary	Uni(T)	Page	XKG($P^T \rightarrow P^V$) + LLM ($P^T \rightarrow P^V$)	One-shot	Single	Single-pass	GPT-4	✗	Ans. Ret. Eff.
D2-ScaleAgent (2026a) *(26.08)*	arXiv'26	✗	✗	Uni(V)	Page, Elem.	ColQwen(P^V) + VLM ($P^V \rightarrow [P^V, E^T]$)	Iterative	Multi	Multi-step	GPT-4o	✗	Ans. Ret. Eff.
DocClaw (2026b) *(26.08)*	arXiv'26	✓	Tree	Mixed(T/V)	Page, Elem.	BM25(P^T; E^T) + ColBERT (P^T; E^T) + ColPali(P^V) + VLM(P^V; E^V)	Iterative	Single	Multi-step	GPT-5/ Gemini	✗	Ans. Eff.
ViSAR (2026) *(26.09)*	arXiv'26	✗	✗	Uni(V)	Page	ColPali/ColQwen/ ColModernVBERT(P^V)	One-shot	Single	Single-pass	Qwen-VL	✗	Ans. Ret. Eff.
CAVR+WSPS+VET (2026) *(26.09)*	arXiv'26	✓	Graph	Mixed(T/V)	Page, Elem.	ColPali(P^V) + WSPS^t(P^V) + VET^t([P^V, E^T])	One-shot	Single	Single-pass	Qwen-VL	SFT	Ans. Ret. Eff.
SCORE (2026c) *(26.09)*	arXiv'26	✗	✗	Uni(V)	Page, Elem.	ColQwen(P^V) + VLM^t(P^V; E^V)	Iterative	Single	Multi-step	Qwen-VL	SFT + RL	Ans. Ret. Eff.
TAEC (2026b) *(26.09)*	arXiv'26	✓	Summary	Mixed(T/V)	Page	Qwen-VL-Embedding(P^V) + Keyword($P^T \rightarrow P^V$) + VLM (P^V)	Iterative	Single	Multi-step	Gemini	✗	Ans. Ret. Eff.

End of Table 6

Header abbreviations. Deep = Deep Parsing; Gran. = Granularity; Acq. = Acquisition; Eval. = Evaluation.

Type labels. Deep and Learning list structure and training types directly; a standalone ✓ in Learning denotes added training without a listed type. SFT/RL: supervised fine-tuning/reinforcement learning.

1. Mechanism. D/P/E denotes document/page/element; superscripts T/V/TV denote text/visual/both. The left side of → is the evidence representation returned by retrieval or selection; the right side is the generation input. Without an arrow, retrieval and generation use the same granularity and modality; retrieval-only methods are labeled by their returned unit. Superscript t denotes additional retrieval, scoring, or page-selection training. Parentheses after model names specify evidence forms; semicolons separate independent paths, and an arrow applies only within its path. Brackets [ ] group multiple evidence items, with commas denoting parallel items.

2. Evaluation. Only dimensions quantitatively reported in the original paper are listed. Ret. evaluates evidence retrieval, usually page-level hits, recall, or ranking; Loc. evaluates within-page element/region localization or coverage, such as IoU or region-overlap recall. **Exceptions (a):** Ret. is document retrieval for DocMMIR, DocPC, NeuSym-RAG, and Agentic Hybrid (EXTRACTOR), document identification for HiKEY, and additionally document recall for HEAVEN; LILaC evaluates layout-element retrieval, and Doc-Researcher evaluates document/page/layout-element retrieval. G^2-Reader Ret. measures semantic coverage of evidence statements; LFRAG includes element-ID matching; HierDoc includes region-ID-set F1; MoDora evaluates evidence-node recall and filtering. ReAlign Loc. measures overlap between attention regions and supervised boxes; RGenCite includes VLM-assessed evidence support; Lang2Act measures the proportion of internal attention falling within gold regions. MMRetHeads evaluates pages and layout regions on MMDocIR; LMS-Retrieval Ret./Loc. measure layout-element recall and region coverage.